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Chapter 8: Predicting What Comes Next - Sequences and Progressions

This chapter trains the single most exam-useful skill in school mathematics: looking at a list of numbers and predicting what comes next WITHOUT writing the whole list. The uploaded PDF moves from simple patterns (odd numbers, triangular numbers, square numbers) to the notation t_n, then to two ways of defining a sequence - the **explicit rule** (a formula in n that jumps straight to any term) and the **recursive rule** (each term built from earlier terms). It then develops the two superstar progressions: the **Arithmetic Progression (AP)** with $t_n = a + (n - 1)d$, the sum of the first n natural numbers $S_n = n(n + 1)/2$ (the Gauss-Āryabhaṭa pairing idea), and the **Geometric Progression (GP)** with $t_n = ar^{n-1}$. Along the way you meet the Virahānka-Fibonacci sequence (born in 7th-century India from Prakrit poetry metres!), the Sierpiński triangle and carpet fractals, bouncing balls, doubling bacteria and taxi fares - all of them secretly sequences. In this practice book every one of those ideas is pushed harder: missing-value reasoning, two-equation attacks, fractal counting, membership tests and HOTS proofs.

8.1 Complete Concept Cheat Sheet (read this before any question)

8.1.1 Sequences, terms and the t-sub-n notation

What is it? A **sequence** is an ordered list of numbers, and each number is called a **term**. Order is sacred: 2, 4, 6, 8 and 4, 2, 8, 6 are different sequences even though they use the same numbers. A sequence may be **infinite** (1, 2, 3, 4, ... - the three dots mean it never ends) or **finite** (6, 12, 24, 48, 96 - exactly five terms). We write t_1 for the first term, t_2 for the second, and so on; the subscript is the **position number**. For several sequences at once we use different letters: t_n, s_n, u_n.

When do I use it? Every single question of this chapter - first identify the terms and their positions. **How do I recognise it?** Any comma-separated list with a '...' or a question about 'the nth term', 'the 20th term', 'which term is...'. **Careful:** n (the position) is always a counting number 1, 2, 3, ... but the term value itself can be negative, a fraction, even irrational - the sequence -7, -3, 1, 5, ... has $s_1 = -7$, and 1, 1/2, 1/3, 1/4, ... has unit fractions as terms.

Tiny example. In the square numbers 1, 4, 9, 16, 25, 36, ... we have $t_1 = 1$, $t_2 = 4$, $t_5 = 25$. '25 is the 5th term' and '$t_5 = 25$' say the same thing.

8.1.2 The famous sequences you must recognise on sight

- **Natural numbers:** 1, 2, 3, 4, ... Each term is one more than the previous one.
- **Odd numbers:** 1, 3, 5, 7, 9, ... Constant difference 2. nth term $t_n = 2n - 1$.
- **Triangular numbers:** 1, 3, 6, 10, 15, ... Each term is the sum 1 + 2 + ... + n, drawn as a triangle of dots. Differences are 2, 3, 4, 5, 6 (increasing, so it is NOT an AP).
- **Square numbers:** 1, 4, 9, 16, 25, ... Each term is the sum of the first n odd numbers: 9 = 1 + 3 + 5. Differences 3, 5, 7, 9 (consecutive odd numbers).
- **Powers of 2:** 2, 4, 8, 16, 32, ... and 1, 2, 4, 8, ... Constant ratio 2 - a GP in disguise.
- **Primes:** 2, 3, 5, 7, 11, 13, ... NO simple rule - a famous example of a sequence without a regular pattern. Examiners love using primes as the 'NOT an AP' option.

8.1.3 Explicit rule - the jump-straight-to-any-term formula

What is it? An **explicit rule** computes the term in position n directly from n, with no need for earlier terms. Examples from the PDF: odd numbers $u_n = 2n - 1$; the sequence $s_n = 5n - 2$ which starts 3, 8, 13, 18, ...; triangular numbers $t_n = n(n + 1)/2$. **Why is it so useful?** Because t_{300} needs ONE substitution, not 299 additions. **How do I use it?** Two directions: (i) substitute n to get the term, (ii) set $t_n = k$ and solve for n to check whether k is a term and find its position.

The membership test (super important). To check if a number k belongs to the sequence, solve $t_n = k$. If n comes out a natural number - k is a term, and n is its position. If n is a fraction or decimal - k is NOT a term, because positions must be 1, 2, 3, ... Example from the PDF: in $s_n = 5n - 2$, solving $5n - 2 = 308$ gives $n = 62$ (a term, the 62nd), while $5n - 2 = 471$ gives $n = 94.6$ (not a term). **Tiny example.** Is 137 an odd number of the list 1, 3, 5, ...? Solve $2n - 1 = 137$, so $n = 69$ - yes, the 69th term.

8.1.4 Recursive rule - each term grows out of the previous ones

What is it? A **recursive rule** defines new terms using earlier terms: you must know the starting value(s) plus the recipe. Examples from the PDF: $t_1 = 1$, $t_n = t_{n-1} + 3$ gives 1, 4, 7, 10, ...; $u_1 = 1$, $u_n = 2u_{n-1} + 3$ gives 1, 5, 13, 29, ...; $s_1 = 3$, $s_n = s_{n-1}(s_{n-1} - 1)$ gives 3, 6, 30, 870 (terms can explode!). A recursion may even use TWO previous terms. **When do I use it?** Whenever the question says 'each term is 3 more than / twice / the previous term(s)' or gives t_1 and a rule. **Method:** iterate step by step, writing every term - never skip, because each step feeds the next.

The Virahānka-Fibonacci sequence (India's gift to mathematics). $V_1 = 1$, $V_2 = 2$, $V_n = V_{n-1} + V_{n-2}$ for n at least 3, giving 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ... Each term is the sum of the previous TWO. Virahānka wrote it down in the 7th century CE in his work Vṛttajātisamuchaya while studying Prakrit poetry metres; Gopāla (c. 1135 CE) and Hemachandra (c. 1150 CE) developed it further, and Fibonacci studied it around 1200 CE - which is why it also carries his name. Memorise the first ten terms: 1, 2, 3, 5, 8, 13, 21, 34, 55, 89 (then 144, 233, ...).

> **Trap alarm**
> The recursive rule and the explicit rule describe the SAME sequence in two languages. '$t_n = t_{n-1} + 3$, $t_1 = 1$' and '$t_n = 3n - 2$' are the same sequence 1, 4, 7, 10, ... Exam questions often ask you to convert one language into the other.

8.1.5 Arithmetic Progression (AP) - the constant-difference sequence

What is it? A sequence in which every term after the first is obtained by ADDING a fixed number d - the **common difference** - to the previous term. General form: a, a + d, a + 2d, a + 3d, ..., a + (n - 1)d, where a is the first term. The nth term is the star formula of this chapter: $\mathbf{t_n = a + (n - 1)d}$. Its recursive form: $t_1 = a$, $t_n = t_{n-1} + d$. Examples: 1, 5, 9, 13 (a = 1, d = 4); 11, 7, 3, -1, -5 (a = 11, d = -4 - a DECREASING AP); 7, 7, 7, ... (d = 0); 1/2, 5/2, 9/2, ... (d = 2).

When do I use it? Words like 'each time 5 more', 'fixed fee plus per-km charge', 'salary increment', 'rows of seats', 'numbers divisible by k'. **How to find d:** $d = t_2 - t_1$, and CHECK it on $t_3 - t_2$ - if the differences differ, it is not an AP. **Why $t_n = a + (n - 1)d$ works:** to reach the nth term you step from a exactly (n - 1) times, each step adding d. **Tiny example.** AP 3, 8, 13, 18: $t_{10} = 3 + 9 \times 5 = 48$. **Plot fact:** plotting the pairs (1, 1), (2, 5),

(3, 9), ... for an AP always lands on a STRAIGHT LINE - the graph of the AP is linear, because the growth per step is constant.

> **Identify-the-method trick**
> Any AP question with TWO unknown facts about two different terms (e.g. '3rd term is 12 and 50th term is 106') becomes two linear equations: a + 2d = 12 and a + 49d = 106. Subtract to get d, back-substitute to get a. This one move solves a huge family of exam questions.

8.1.6 Sum of the first n natural numbers - the Gauss-Āryabhaṭa formula

What is it? $S_n = 1 + 2 + 3 + ... + n = n(n + 1)/2$. The beautiful proof: write S forwards and backwards, add - each column pairs to (n + 1) and there are n columns, so 2S = n(n + 1). Āryabhaṭa's Āryabhaṭīya (Chapter 2, Verse 19) records this idea centuries earlier: take (first + last)/2 - the average - and multiply by the number of terms. The same average trick sums ANY equally spaced list: sum = (first + last)/2 × count.

Range sums: 25 + 26 + ... + 58 = (1 + 2 + ... + 58) - (1 + 2 + ... + 24) = $S_{58} - S_{24}$ = 29 × 59 - 12 × 25 = 1711 - 300 = 1411. Cut off the unwanted head of the list! **Triangular numbers:** the nth triangular number IS S_n, so $t_{10} = 55$, $t_{17} = 153$, $t_{80} = 80 \times 81/2 = 3240$. **Tiny example.** 1 + 2 + ... + 100 = 100 × 101/2 = 5050 - Gauss's legendary schoolboy answer.

8.1.7 Geometric Progression (GP) - the constant-ratio sequence

What is it? A sequence in which every term after the first is obtained by MULTIPLYING the previous term by a fixed number r - the **common ratio**. General form: a, ar, ar^2, ar^3, ..., ar^{n-1}, with nth term $\mathbf{t_n = ar^{n-1}}$ and recursive rule $t_1 = a$, $t_n = r \cdot t_{n-1}$. Check by division: $t_2/t_1 = t_3/t_2 = ... = r$. Examples: 3, 6, 12, 24 (r = 2); 1, -1, 1, -1 (r = -1 - alternating); 5, 15/4, 45/16, ... (r = 3/4 - shrinking); 1, 1/3, 1/9, ... (r = 1/3).

When do I use it? Keywords: 'doubles', 'triples', 'halves each time', 'bounces to 3/4 of its height', 'grows 10% per year', 'bacteria multiply', 'fractal splits into 3'. Fractional and negative ratios are full citizens: r = 3/4 makes the sequence shrink towards 0 (decay), r = -1 makes it oscillate. **Plot fact:** a GP does NOT plot on a straight line - the points (1, 3), (2, 6), (3, 12), (4, 24) curve upwards ever more steeply (exponential growth), or melt towards zero when 0 < r < 1. **Tiny example.** 3, 6, 12, 24, 48: $t_7 = 3 \times 2^6 = 192$.

8.1.8 Fractals and real-life GPs (Sierpiński, bacteria, bouncing balls)

- **Sierpiński triangle:** Stage 0 is one black triangle. Each stage replaces every black triangle by 3 smaller ones (join midpoints, remove the centre). Black triangle count: 1, 3, 9, 27, 81, ... = 3^n at Stage n (a GP with a = 1, r = 3 - note the indexing starts at Stage 0!). Black AREA (Stage 0 = 1 square unit): 1, 3/4, $(3/4)^2$, ... = $(3/4)^n$ - a shrinking GP that heads to 0. Count explodes, area melts: that is the fractal magic.
- **Sierpiński square carpet:** trisect each side, remove the middle ninth. Red squares: 8^n at Stage n; red area: $(8/9)^n$.
- **Bouncing ball:** dropped from 24 ft, each bounce rises to 3/4 of the previous height: 18, 13.5, 10.125, ... - GP with a = 18, r = 3/4.
- **Bacteria doubling:** 30 bacteria double every hour: 30×2^n after n hours - growth GP.
- **Fractals in nature:** trees, cauliflower, broccoli, snowflakes, coastlines - shapes that repeat themselves at different scales; zoom in and you see the whole again.

> **Indexing trap of the year**
> Sierpiński counting starts at Stage ZERO: Stage 0 has 1 triangle, so Stage n has 3^n (not 3^{n-1}). Bounce questions are the opposite: bounce 1 is the FIRST bounce. Underline in the question whether stages start at 0 or 1 before you touch the formula.

8.1.9 Exam tips for this chapter

1. Before anything, decide the species: constant ADDITION means AP, constant MULTIPLICATION means GP, neither means look for a positional rule like n^2 or a recursive recipe.
2. Compute d from $t_2 - t_1$ and VERIFY with $t_3 - t_2$; compute $r = t_2/t_1$ and verify with t_3/t_2. One verification saves whole marks.
3. 'Is k a term?' always means: solve $t_n = k$, then check that n is a natural number. A fractional n is a polite NO.
4. Far terms (50th, 100th, 1000th) demand the explicit formula - NEVER list terms one by one.
5. Two facts about two terms means two equations in a and d (AP) or a and r (GP). Subtract/divide to eliminate.
6. For sums $1 + 2 + \ldots + n$ use $n(n + 1)/2$; for sums starting mid-list subtract two S-formulas, or use (first + last)/2 × count.
7. Watch the indexing: stages of Sierpiński start at 0 (count 3^n), bounces start at 1 (height $a \cdot r^{\text{bounce}}$).
8. Sanity-check the type of growth: if answers explode, $r > 1$; if they melt to 0, $0 < r < 1$; if they go down steadily, $d < 0$.

8.2 Question Types / Concept Map for Chapter 8

Almost every sequences-and-progressions question in exams falls into one of the following types. Label the question before solving - knowing WHICH fight it is wins half the battle.

Type	Name	How to recognise it in the question	Core method
T1	Sequence vocabulary and notation	'term', 'finite/infinite', 'state t_5', position questions	Read positions; subscript = position
T2	List terms from a rule	t_n given, 'find the first k terms'	Substitute n = 1, 2, 3, ... one by one
T3	Far-term jump	'find the 100th term', 'the 1170th term'	One substitution into the explicit rule
T4	Membership check	'is k a term? which term is k?'	Solve $t_n = k$; n must be a natural number
T5	Recursive generation	t_1 given, t_n in terms of t_{n-1} (or two back)	Iterate term by term, no skipping
T6	Virahānka-Fibonacci	'each term = sum of previous two', poetry metres, rabbit style	Add last two; know 1, 2, 3, 5, 8, 13, ...
T7	AP identify and d	'constant difference', decreasing lists	$d = t_2 - t_1$, verify; d may be negative or 0

Type	Name	How to recognise it in the question	Core method
T8	AP nth term / which term	'find t_{20}', 'which term is 607?'	$t_n = a + (n - 1)d$; or solve for n
T9	AP from two conditions	'3rd term is 12, 50th is 106'	Two linear equations in a and d
T10	Sums of naturals and ranges	1 + 2 + ... + n; k + ... + m; smallest n with $S_n >$ M	$n(n + 1)/2$; $S_{big} - S_{small}$; average × count
T11	GP identify and ratio	'doubles', 'halves', alternating signs	$r = t_2/t_1$, verify; r may be negative or fractional
T12	GP nth term / which term	'find t_{10}', 'which term is 4374?'	$t_n = ar^{n-1}$; powers of r; handle fractions
T13	Fractals and real-life GPs	Sierpiński stages, bacteria, bouncing balls, 10% growth	Set up a and r; fix the stage/bounce indexing
T14	Plot nature and HOTS mix	'do the points lie on a line?', proofs, hybrid stories	AP plots straight; GP curves; split into mini-steps

Distribution of the 120 practice questions below: T1-T5 feed the Warm-up Zone (Q1-Q32), T6-T12 build the Core Drills (Q33-Q76), T9-T13 power the Exam Power Zone (Q77-Q104), and T14 with multi-concept HOTS rules the Boss Zone (Q105-Q120).

8.3 Speed Tricks & Trap Alarms

Ye table tumhara "dekho aur maaro" radar hai - question ka trigger phrase dikha, toh yahan ka trick fire karo. Inhe ratt lo, exam me sochna kam padega aur speed double ho jayegi.

Instant trick	Fire it when...
Constant +/- between terms? It is an AP: $d = t_2 - t_1$, $t_n = a + (n - 1)d$. 10th term = a + 9d	'Each time 5 more', fixed fee + per-km charges, yearly increments, row/seat counts
Constant x or divide between terms? It is a GP: $r = t_2/t_1$, $t_n = ar^{n-1}$. t_{10} of 3, 6, 12 = 3 x 2^9 = 1536	'Doubles', 'halves', '3/4 of previous height', '10% growth', 'splits into 3'
'Is k a term?' - solve $t_n = k$. n a natural number = YES at position n; n fractional = NO	'Is 100 a term?', 'which term is 607?', 'prove 461 is not a term'
Two term-facts given? Two equations: $a + (n - 1)d$ = value (AP) or ar^k = value (GP). Subtract / divide them	'3rd term is 12 and 50th is 106', '$t_{11} = 38$, $t_{16} = 73$', '$t_2 = 9$, $t_5 = 243$'
Sums: $1 + ... + n = n(n + 1)/2$. Range $k + ... + m = S_m - S_{k-1}$, or (first + last)/2 x count	'Total marbles', 'sum of all multiples of 3 up to 99', '25 + 26 + ... + 58'
Real-life GP: the exponent counts the doublings / stages / bounces. Sierpinski starts at Stage 0, so count = 3^n there	'Bacteria double every hour', 'bounces to 3/4', 'Stage 6 of the fractal', 'folded 20 times'
Graph fingerprint: AP points lie on a straight line of slope d; GP points bend on a steepening curve ($r > 1$) or melt to 0 ($0 < r < 1$)	'Do the points lie on a line?', 'what shape does the plot make?', 'sketch the growth'

8.4 The 120-Question Workout

Poora chapter ab ek hi ring mein: **120 questions**, chaar level zones mein banta hua. Warm-up se shuru karo, Boss Zone tak pahuncho. Har question ke turant baad boxed step-by-step solution hai - pehle khud try karo, atako toh solution dekho, aur "Why this step?" + Hinglish note se logic pakka karo. Agar in 120 ko solve kar liya toh is chapter ka koi bhi twisted question tumhe nahi rok payega.

Level 1 · Warm-up Zone: basics and confidence:

QUESTION 1 | Basic | Sequence terms and notation (T1)

Q1. For the sequence of square numbers 1, 4, 9, 16, 25, 36, ... write t_1, t_2, t_5 and t_6. Is this sequence finite or infinite? What is the position number of 144 if the pattern continues?

SOLUTION — step by step

Step 1. A sequence is an ordered list; the subscript of t tells the POSITION of the term.

Step 2. Reading positions: $t_1 = 1$, $t_2 = 4$, $t_5 = 25$, $t_6 = 36$.

Step 3. The three dots ... mean the list never ends, so the sequence is infinite.

Step 4. $144 = 12^2$ is the square of 12, so it sits at position n = 12: $t_{12} = 144$.

» **Why this step?** The subscript is an address, not a value: $t_5 = 25$ says 'at position 5 the value is 25'.

Answer: $t_1 = 1$, $t_2 = 4$, $t_5 = 25$, $t_6 = 36$; infinite; 144 is the 12th term.

Hinglish: *Sequence matlab line mein laghi hui numbers - position fix hai. t_5 poochha toh 5th position wala number dena hai. Ellipsis (...) dikhe toh samjho list kabhi khatam nahi hoti - infinite.*

QUESTION 2 | Basic | Finite vs infinite sequences (T1)

Q2. Classify each sequence as finite or infinite, giving the number of terms when finite: (i) 1, 2, 3, 4, ... (ii) 6, 12, 24, 48, 96 (iii) 1, 1/2, 1/3, 1/4, ... (iv) the page numbers 1 to 250 of a book listed in order.

SOLUTION — step by step

Step 1. (i) The dots ... signal that the list continues forever - infinite (countably endless).

Step 2. (ii) The list stops at 96 - finite, and counting gives exactly 5 terms.

Step 3. (iii) Unit fractions continue as 1/5, 1/6, 1/7, ... forever - infinite (the terms shrink towards 0 but never end).

Step 4. (iv) Page numbers run 1, 2, 3, ..., 250 and stop - finite with exactly 250 terms.

» **Why this step?** Finite/infinite is decided ONLY by whether the list stops - the size of the terms is irrelevant.

Answer: (i) infinite (ii) finite, 5 terms (iii) infinite (iv) finite, 250 terms

Hinglish: *Three dots (...) = hamesha chalta rahega = infinite. List kahin ruk gayi = finite, terms gino lo. Page numbers jaise real-life lists finite hoti hain; unit fractions ki list kabhi nahi rukti kyunki har n ke liye 1/n banta hai.*

QUESTION 3 | Basic | Next terms of famous sequences (T1/T6)

Q3. Write the next three terms with the reason for each pattern: (i) 1, 3, 5, 7, ... (ii) 1, 4, 9, 16, ... (iii) 1, 3, 6, 10, ... (iv) 2, 4, 8, 16, ...

SOLUTION — step by step

Step 1. (i) Odd numbers: each term +2. Next: 9, 11, 13.

Step 2. (ii) Square numbers: 1, 4, 9, 16 are 1^2, 2^2, 3^2, 4^2. Next: 5^2, 6^2, 7^2 = 25, 36, 49. (Differences 3, 5, 7, 9, 11 - consecutive odd numbers.)

Step 3. (iii) Triangular numbers: 1, 1+2, 1+2+3, ... Each step adds one more than before: +2, +3, +4, so next +5, +6, +7 gives 15, 21, 28.

Step 4. (iv) Powers of 2: each term is doubled (ratio 2). Next: 32, 64, 128.

» **Why this step?** Same four numbers, four different rules - always state the RULE, not just the next numbers.

Answer: (i) 9, 11, 13 (ii) 25, 36, 49 (iii) 15, 21, 28 (iv) 32, 64, 128

Hinglish: *Pattern dekho: +2 wale odd, square wale perfect squares, triangular wale mein gap badhta jaata hai (+2, +3, +4...), aur 2, 4, 8 mein har baar double. Reason likhna zaroori hai - sirf answer likha toh aadha mark bhi nahi milta.*

QUESTION 4 | Basic | Triangular numbers as sums (T1/T10)

Q4. Write the 5th, 6th, 7th and 8th triangular numbers, each expressed as a sum of natural numbers. Then verify that the 8th triangular number equals 8 × 9/2.

SOLUTION — step by step

Step 1. The nth triangular number is 1 + 2 + ... + n.

Step 2. $T_5 = 1 + 2 + 3 + 4 + 5 = 15$.

Step 3. $T_6 = 15 + 6 = 21 = 1 + 2 + 3 + 4 + 5 + 6$.

Step 4. $T_7 = 21 + 7 = 28$; $T_8 = 28 + 8 = 36$.

Step 5. Formula check: $T_8 = 8 \times 9/2 = 72/2 = 36$. ✓

» **Why this step?** Each new triangular number adds the next natural number - that is why $T_n = n(n + 1)/2$.

Answer: $T_5 = 15$, $T_6 = 21$, $T_7 = 28$, $T_8 = 36$; formula verified.

Hinglish: *Triangular number = 1 se n tak ki sum. Pichhle term mein agla number jodo - 15 + 6 = 21, 21 + 7 = 28. Formula n(n+1)/2 se seedha 36 mil gaya - dono raaste same jagah pahunchte hain.*

QUESTION 5 | Basic | Squares as sums of odd numbers (T1/T10)

Q5. Express 9 and 16 as sums of consecutive odd numbers starting from 1. Then find 1 + 3 + 5 + 7 + 9 without adding one by one, and state the general rule.

SOLUTION — step by step

Step 1. 9 = 1 + 3 + 5 (first 3 odd numbers) and $3^2 = 9$. ✓

Step 2. 16 = 1 + 3 + 5 + 7 (first 4 odd numbers) and $4^2 = 16$. ✓

Step 3. 1 + 3 + 5 + 7 + 9 is the sum of the first 5 odd numbers, so it must be $5^2 = 25$.

Step 4. General rule: sum of the first n odd numbers = n^2.

» **Why this step?** The square pattern 1, 4, 9, 16 grows by consecutive odd numbers 3, 5, 7, ... - each new odd number builds the next L-shaped layer of the square.

Answer: 9 = 1 + 3 + 5; 16 = 1 + 3 + 5 + 7; 1 + 3 + 5 + 7 + 9 = 25; sum of first n odd numbers = n^2.

Hinglish: *Square number ka GUI hai odd numbers ka sum. 5 odd numbers ka sum poochha? Seedha 5 ka square = 25. Ye identity exam mein 'without adding' wale questions ki jaan hai.*

QUESTION 6 | Basic | Listing terms from an explicit rule (T2)

Q6. A sequence is given by the explicit rule $t_n = 3n + 1$ for n at least 1. Write its first five terms. Is this sequence increasing or decreasing?

SOLUTION — step by step

Step 1. Substitute n = 1, 2, 3, 4, 5 one by one.

Step 2. $t_1 = 3(1) + 1 = 4$.

Step 3. $t_2 = 3(2) + 1 = 7$; $t_3 = 3(3) + 1 = 10$.

Step 4. $t_4 = 3(4) + 1 = 13$; $t_5 = 3(5) + 1 = 16$.

Step 5. Each term is 3 more than the one before it, so the sequence increases (it is an AP with d = 3).

» **Why this step?** An explicit rule is a machine: feed the position n, get the term. No earlier term is needed.

Answer: 4, 7, 10, 13, 16 - increasing (an AP with d = 3).

Hinglish: *Explicit formula = seedha substitiution machine. n = 1, 2, 3, 4, 5 daalo, terms nikal aaye: 4, 7, 10, 13, 16. Gap har jagah 3 - matlab ye AP bhi hai, common difference 3.*

QUESTION 7 | Basic | Far terms of the odd numbers (T3)

Q7. The odd numbers have the explicit rule $u_n = 2n - 1$. Verify u_1 to u_4, then find the 20th, 53rd, 108th and 1170th terms WITHOUT listing any terms.

SOLUTION — step by step

Step 1. Check: $u_1 = 2(1) - 1 = 1$, $u_2 = 3$, $u_3 = 5$, $u_4 = 7$ - exactly the odd numbers. ✓

Step 2. $u_{20} = 2 \times 20 - 1 = 39$.

Step 3. $u_{53} = 2 \times 53 - 1 = 105$.

Step 4. $u_{108} = 2 \times 108 - 1 = 215$.

Step 5. $u_{1170} = 2 \times 1170 - 1 = 2339$.

» **Why this step?** This is the whole point of an explicit rule: the 1170th term costs ONE multiplication and ONE subtraction - no listing, ever.

Answer: $u_1 ... u_4 = 1, 3, 5, 7$; $u_{20} = 39$, $u_{53} = 105$, $u_{108} = 215$, $u_{1170} = 2339$.

Hinglish: *Ek bhi term likhe bina 1170th term nikal di - bas formula mein n daala. Ye hi toh explicit rule ki taakat hai. Exam mein 'find the 100th term' dikhe toh listing karne wala student 100 lines likhega, tum 1 line.*

QUESTION 8 | Basic | Generating a sequence from s-sub-n (T2/T3)

Q8. Consider the sequence generated by $s_n = 5n - 2$. (i) Write the first six terms. (ii) Find the 100th term. (iii) Find the 1000th term.

SOLUTION — step by step

Step 1. $s_1 = 5(1) - 2 = 3$; $s_2 = 8$; $s_3 = 13$; $s_4 = 18$; $s_5 = 23$; $s_6 = 28$.

Step 2. The first six terms are 3, 8, 13, 18, 23, 28 (an AP with a = 3, d = 5).

Step 3. $s_{100} = 5 \times 100 - 2 = 498$.

Step 4. $s_{1000} = 5 \times 1000 - 2 = 4998$.

» **Why this step?** Notice the structure: first term 3 and each step +5 - so $s_n = 3 + (n - 1) \times 5 = 5n - 2$. The explicit rule and the AP picture agree.

Answer: (i) 3, 8, 13, 18, 23, 28 (ii) 498 (iii) 4998

Hinglish: *Terms banaye toh 3 se shuru, har baar +5. 100th term ke liye formula: 500 - 2 = 498. Listing karne wala yahan 100 terms likhta - tumne 2 second mein kaam khatam.*

QUESTION 9 | Basic | Substitution practice (T2/T3)

Q9. For the sequence $t_n = 3n - 7$, find (i) the first, second and third terms, (ii) the 12th, 18th and 50th terms, (iii) which term of the sequence is 332.

SOLUTION — step by step

Step 1. (i) $t_1 = 3 - 7 = -4$; $t_2 = 6 - 7 = -1$; $t_3 = 9 - 7 = 2$. The sequence starts -4, -1, 2 (each step +3).

Step 2. (ii) $t_{12} = 36 - 7 = 29$; $t_{18} = 54 - 7 = 47$; $t_{50} = 150 - 7 = 143$.

Step 3. (iii) Set $3n - 7 = 332$, so $3n = 339$, giving $n = 113$.

Step 4. n = 113 is a natural number, so 332 is the 113th term.

» **Why this step?** Part (iii) runs the rule BACKWARDS: instead of n giving the term, the term gives the position n.

Answer: (i) -4, -1, 2 (ii) 29, 47, 143 (iii) the 113th term

Hinglish: *Negative terms se mat ghabrao - formula kya bole wahi likho. Pehle teen terms -4, -1, 2 hain (AP, d = 3). 332 kaunsa term hai? 3n = 339, n = 113 - hazaar term door bhi formula ek second mein pahuncha deta hai.*

QUESTION 10 | Basic | Membership test (T4)

Q10. For the sequence $t_n = 3n - 7$, check whether 557 is a term. Also check whether 460 is a term. Justify both answers.

SOLUTION — step by step

Step 1. Test 557: solve $3n - 7 = 557$, so $3n = 564$ and $n = 564/3 = 188$.

Step 2. n = 188 is a natural number, so 557 IS a term - exactly the 188th.

Step 3. Test 460: solve $3n - 7 = 460$, so $3n = 467$ and $n = 467/3 = 155.666...$

Step 4. n is not a whole number, so 460 is NOT a term of this sequence.

» **Why this step?** The membership test: k is a term if and only if (k + 7)/3 lands on a natural number. A decimal n is impossible because positions are 1, 2, 3, ... only.

Answer: 557 is a term (the 188th); 460 is not a term (n = 467/3 is not natural).

Hinglish: *Membership test ka funda: equation solve karo, n whole number aaya toh haan, fraction aaya toh na. 557 ke liye n = 188 perfect. 460 ke liye n = 155.67 - aisa position exist hi nahi karta. Bas yahi logic exam mein 2 mark dilaata hai.*

QUESTION 11 | Basic | Decreasing sequence from a rule (T2/T7)

Q11. Write the first five terms of the sequence $t_n = 2 - 5n$. Does it increase or decrease? Find its common difference if it is an AP.

SOLUTION — step by step

Step 1. $t_1 = 2 - 5(1) = -3$; $t_2 = 2 - 10 = -8$; $t_3 = 2 - 15 = -13$.

Step 2. $t_4 = 2 - 20 = -18$; $t_5 = 2 - 25 = -23$.

Step 3. First five terms: -3, -8, -13, -18, -23.

Step 4. Each term is 5 LESS than the previous one, so the sequence decreases; the constant gap is d = -5 (it is a decreasing AP).

» **Why this step?** The coefficient of n controls the direction: +5n climbs, -5n falls. A negative common difference is perfectly legal.

Answer: -3, -8, -13, -18, -23; decreasing; it is an AP with d = -5.

Hinglish: *Minus sign ne sequence ko ulta kar diya - terms har baar 5 girti hain. AP ho ya na ho, yeh check karne ka tareeka: consecutive gaps same hain? Hain (-5) - toh decreasing AP hai, d = -5.*

QUESTION 12 | Basic | Quadratic rule is not an AP (T2/T14)

Q12. Find the first five terms of the sequence $t_n = n^2 - 2n + 3$. Are the differences between consecutive terms constant? What does that tell you?

SOLUTION — step by step

Step 1. $t_1 = 1 - 2 + 3 = 2$; $t_2 = 4 - 4 + 3 = 3$; $t_3 = 9 - 6 + 3 = 6$.

Step 2. $t_4 = 16 - 8 + 3 = 11$; $t_5 = 25 - 10 + 3 = 18$.

Step 3. Terms: 2, 3, 6, 11, 18. Differences: 3 - 2 = 1, 6 - 3 = 3, 11 - 6 = 5, 18 - 11 = 7.

Step 4. The differences 1, 3, 5, 7 are NOT constant (they are consecutive odd numbers), so this is NOT an AP - it is a quadratic sequence.

» **Why this step?** When the rule for t_n contains n^2, the differences grow in a straight-line pattern themselves - the signature of a quadratic sequence.

Answer: 2, 3, 6, 11, 18; differences 1, 3, 5, 7 are not constant; not an AP.

Hinglish: *n square wala rule = differences apne aap odd numbers ban jaate hain: 1, 3, 5, 7. Isliye ye AP nahi hai. Dekha? Sequence ka 'speed' bhi pattern follow karti hai - isi ko second difference kehte hain, aage ke classes mein yehi bahut kaam aayega.*

QUESTION 13 | Basic | Unit fractions (T1/T2)

Q13. The sequence 1, 1/2, 1/3, 1/4, ... has $t_1 = 1$ and unit fractions after it. Write t_5, t_6, t_7, t_8, give the explicit rule, and state whether the sequence increases or decreases.

SOLUTION — step by step

Step 1. The term at position n is 1/n: $t_5 = 1/5$, $t_6 = 1/6$, $t_7 = 1/7$, $t_8 = 1/8$.

Step 2. Explicit rule: $t_n = 1/n$.

Step 3. As n grows, the denominator grows, so each fraction gets smaller: $1/5 > 1/6 > 1/7 > 1/8$.

Step 4. The sequence strictly decreases, sliding towards 0 (but never reaching it).

» **Why this step?** A sequence can decrease and still stay positive forever - it never actually touches 0, it only gets arbitrarily close.

Answer: $t_5 = 1/5$, $t_6 = 1/6$, $t_7 = 1/7$, $t_8 = 1/8$; $t_n = 1/n$; strictly decreasing towards 0.

Hinglish: *Yahan har term chhoti hoti jaati hai - 1/5, 1/6, 1/7... Zero ke paas paas jaati hai par kabhi zero nahi hoti. Decreasing ka matlab negative hona nahi hota - values bas girti jaati hain.*

QUESTION 14 | Basic | Sequence with a negative start (T2/T8)

Q14. A sequence starts $s_1 = -7$ and every term after that is 4 more than the previous one. Write the first five terms, find the explicit rule s_n, and find s_{10}.

SOLUTION — step by step

Step 1. Stepping by +4: -7, -3, 1, 5, 9 - the first five terms (the PDF uses exactly this sequence).

Step 2. This is an AP with first term a = -7 and common difference d = 4.

Step 3. Explicit rule: $s_n = a + (n - 1)d = -7 + (n - 1) \times 4 = 4n - 11$.

Step 4. Check: $s_1 = 4 - 11 = -7$ ✓, $s_2 = 8 - 11 = -3$ ✓.

Step 5. $s_{10} = 4 \times 10 - 11 = 29$.

» **Why this step?** A negative first term changes nothing structurally: a + (n - 1)d works for any a, positive or negative.

Answer: -7, -3, 1, 5, 9; $s_n = 4n - 11$; $s_{10} = 29$.

Hinglish: *Minus 7 se shuru hokar bhi sequence ko +4 ki speed mili toh woh seedhi chadne lagi. Formula wahi purana a + (n-1)d. Note karo: 4n - 11 - coefficient of n hi d batata hai.*

QUESTION 15 | Basic | Recursive rule iteration (T5)

Q15. A sequence is given by the recursive rule $u_1 = 1$, $u_n = 2u_{n-1} + 3$ for n at least 2. Find the first four terms. Continuing the pattern, is 133 a term of this sequence?

SOLUTION — step by step

Step 1. $u_1 = 1$ (given).

Step 2. $u_2 = 2u_1 + 3 = 2 \times 1 + 3 = 5$.

Step 3. $u_3 = 2u_2 + 3 = 2 \times 5 + 3 = 13$.

Step 4. $u_4 = 2u_3 + 3 = 2 \times 13 + 3 = 29$. First four terms: 1, 5, 13, 29.

Step 5. Keep going: $u_5 = 2 \times 29 + 3 = 61$, $u_6 = 2 \times 61 + 3 = 125$, $u_7 = 2 \times 125 + 3 = 253$.

Step 6. 133 is not in the list 1, 5, 13, 29, 61, 125, 253, ... - the sequence jumps straight from 125 to 253, skipping 133. So 133 is NOT a term.

» **Why this step?** A recursive rule needs EVERY previous term: you cannot compute u_5 without u_4. Skipping a step breaks the chain.

Answer: 1, 5, 13, 29; no - 133 is not a term (the sequence jumps 125 to 253).

Hinglish: *Recursive rule = har naya term pichhle term ka bachcha. Ek bhi step skip kiya toh poora chain galat. 133 check karne ke liye terms banani padi - 125 ke baad seedha 253, toh 133 list mein nahi hai.*

QUESTION 16 | Basic | Recursive rule with multiplication (T5)

Q16. Find the first four terms of the sequence defined by $s_1 = 3$, $s_n = s_{n-1}(s_{n-1} - 1)$ for n at least 2. What do you notice about how fast the terms grow?

SOLUTION — step by step

Step 1. $s_1 = 3$ (given).

Step 2. $s_2 = s_1(s_1 - 1) = 3 \times 2 = 6$.

Step 3. $s_3 = s_2(s_2 - 1) = 6 \times 5 = 30$.

Step 4. $s_4 = s_3(s_3 - 1) = 30 \times 29 = 870$.

Step 5. First four terms: 3, 6, 30, 870 - the terms explode (3 to 870 in three steps!).

» **Why this step?** When the rule multiplies the previous term by something close to itself, growth is brutal - far faster than any GP with a small fixed ratio.

Answer: 3, 6, 30, 870; growth is explosive (each term is roughly its own square).

Hinglish: *Yahan rule khud ka number khud se multiply karwa raha hai - isliye terms bomb ki tarah phatte hain: 3, 6, 30, 870. Recursive rules ki speed unpredictable hoti hai - isliye aise questions mein terms honestly banani padti hain.*

QUESTION 17 | Basic | Virahanka-Fibonacci sequence (T5/T6)

Q17. The Virahānka-Fibonacci sequence is defined by $V_1 = 1$, $V_2 = 2$, and $V_n = V_{n-1} + V_{n-2}$ for n at least 3, giving 1, 2, 3, 5, 8, 13, 21, 34, ... Write the next two terms and state V_9 and V_{10}. Who studied this sequence in 7th-century India and in what context?

SOLUTION — step by step

Step 1. Each term is the sum of the previous TWO terms.

Step 2. $V_9 = V_8 + V_7 = 34 + 21 = 55$.

Step 3. $V_{10} = V_9 + V_8 = 55 + 34 = 89$. So the next two terms after 34 are 55 and 89.

Step 4. History: Virahānka wrote it down in the 7th century CE in his work Vṛttajātisamuchaya, while studying Prakrit poetry metres; Gopāla (c. 1135) and Hemachandra (c. 1150) developed it further; Fibonacci studied it around 1200 CE.

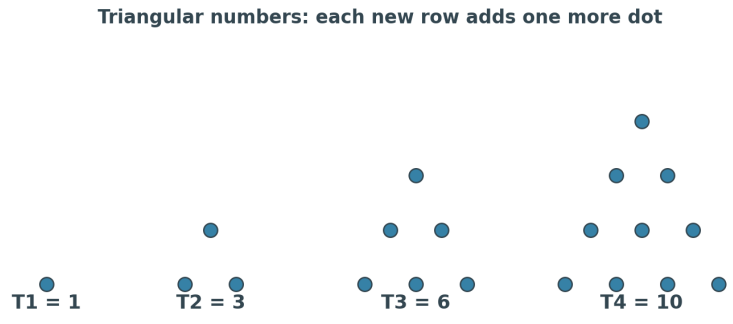
» Why this step? The recursion uses TWO previous terms - that is why you need two starting values (1 and 2) instead of one.

Answer: Next two terms: 55 and 89; $V_9 = 55$, $V_{10} = 89$; Virahānka, 7th century CE, in the study of Prakrit poetry metres.

Hinglish: *Do pichhle terms ka sum = naya term. 21 + 34 = 55, 34 + 55 = 89. Aur history bhi yaad rakho - Virahānka ne 7th century mein Prakrit kavita ke chhand padhte hue discover kiya tha. Fibonacci se pehle India mein ye sequence zinda tha!*

QUESTION 18 | Basic | Triangular dot patterns (T1/T14)

Q18. The figure shows triangular dot arrangements for 1, 3, 6 and 10. Count the dots in each, describe how the pattern grows, and find the 6th triangular number. How does the picture prove that 1 + 2 + ... + 6 = 21?



SOLUTION — step by step

Step 1. Counting rows: 1 dot; 1 + 2 = 3 dots; 1 + 2 + 3 = 6 dots; 1 + 2 + 3 + 4 = 10 dots.

Step 2. Growth: each new row at the bottom adds one dot more than the previous row added (+2, +3, +4, ...).

Step 3. The 6th arrangement adds a row of 6: 1 + 2 + 3 + 4 + 5 + 6 = 21.

Step 4. Picture proof: take two copies of the 6-row triangle; they fit together into a 7 × 6 rectangle of dots (21 + 21 = 42 = 7 × 6). So one triangle is 7 × 6/2 = 21.

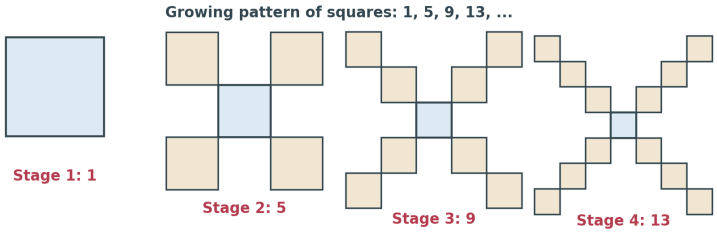
» **Why this step?** Two identical staircases make a rectangle - that is the geometric heart of the formula n(n + 1)/2.

Answer: 1, 3, 6, 10; each row adds one more dot; 6th triangular number = 21; two triangles form a 7 × 6 rectangle, so the sum is 42/2 = 21.

Hinglish: *Har nayi row ek dot zyada lagati hai - isliye 1, 3, 6, 10, 15, 21. Do triangles jodo toh rectangle ban jaata hai: 7 × 6. Aadha rectangle = triangle = 21. Picture dekh ke formula yaad ho jaata hai - rattne ki zaroorat nahi.*

QUESTION 19 | Basic | Growing square pattern (T7/T13)

Q19. The figure shows a growing pattern of squares at Stages 1 to 4, giving the counts 1, 5, 9, 13. How many squares are added at each stage? Predict the counts at Stages 5 and 6, and find the number of squares at Stage 20.



SOLUTION — step by step

Step 1. Differences: 5 - 1 = 4, 9 - 5 = 4, 13 - 9 = 4 - each stage adds exactly 4 new squares at the corners.

Step 2. Stage 5 = 13 + 4 = 17; Stage 6 = 17 + 4 = 21.

Step 3. Constant difference 4 means the counts form an AP: a = 1, d = 4.

Step 4. $t_n = 1 + (n - 1) \times 4 = 4n - 3$.

Step 5. Stage 20: $t_{20} = 4 \times 20 - 3 = 77$.

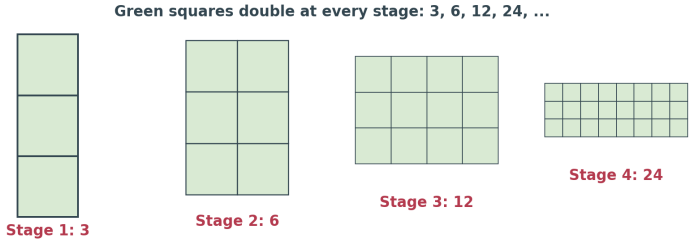
» **Why this step?** When a picture adds the SAME number of objects every stage, the counts must be an AP - the visual pattern becomes the formula 4n - 3.

Answer: 4 squares added each stage; Stage 5 = 17, Stage 6 = 21; Stage 20 = 77 ($t_n = 4n - 3$).

Hinglish: *Har stage par 4 naye squares - bas yahi constant difference hai. Stage 20 ka jawab listing se nahi, formula se aaya: 4n - 3, n = 20 daala, 77. Pattern dekho, AP pakdo, formula likho - yehi asli skill hai.*

QUESTION 20 | Basic | Doubling square pattern (T11/T13)

Q20. A different growing pattern of green squares gives the counts 3, 6, 12, 24 at Stages 1 to 4. Predict Stages 5 and 6. How is this growth different from the 1, 5, 9, 13 pattern of the previous question?



SOLUTION — step by step

Step 1. Ratios: 6/3 = 2, 12/6 = 2, 24/12 = 2 - every stage DOUBLES the previous count.

Step 2. Stage 5 = 24 × 2 = 48; Stage 6 = 48 × 2 = 96.

Step 3. Equal ratios (not equal differences) make this a GP: a = 3, r = 2, so $t_n = 3 \times 2^{n-1}$.

Step 4. Difference from the other pattern: there the counts grew by adding a fixed 4 (AP, straight-line growth); here they grow by multiplying by 2 (GP, explosive growth). Stage 20 here is 3×2^{19} = 1,572,864 - versus just 77 for the AP pattern!

» **Why this step?** Check ratios first, differences second: same-looking lists can be AP or GP, and the two grow at totally different speeds.

Answer: Stage 5 = 48, Stage 6 = 96; it is a GP (r = 2) growing by multiplication, not by the fixed addition of an AP - far faster growth.

Hinglish: *Pichhle question mein plus 4 tha, yahan multiply by 2 hai. Do stages aage = 48, 96. Aur Stage 20? 3 × 2^{19} = 15 lakh se zyada! AP seedhi sadak hai, GP rocket - bas yahi farak samajhna hai.*

QUESTION 21 | Basic | Classify AP / GP / neither (T7/T11)

Q21. Classify each sequence as an AP, a GP, or neither, giving d or r where they exist: (i) 7, 10, 13, 16, ... (ii) 3, 6, 12, 24, ... (iii) 1, 4, 9, 16, ... (iv) 40, 20, 10, 5, ... (v) 2, 3, 5, 7, 11, ...

SOLUTION — step by step

Step 1. (i) Differences 3, 3, 3 - constant: AP with d = 3.

Step 2. (ii) Ratios 2, 2, 2 - constant: GP with r = 2.

Step 3. (iii) Differences 3, 5, 7 - growing; ratios 4, 9/4, 16/9 - not constant either: neither (it is the square sequence).

Step 4. (iv) Differences -20, -10, -5 (halving); ratios 1/2, 1/2, 1/2 - constant: GP with r = 1/2.

Step 5. (v) Consecutive primes: gaps 1, 2, 2, 4 - irregular: neither (no simple rule).

» **Why this step?** Test differences for AP, test ratios for GP. If both fail, hunt for a positional rule (n^2, primes, ...) or admit 'neither'.

Answer: (i) AP, d = 3 (ii) GP, r = 2 (iii) neither (iv) GP, r = 1/2 (v) neither

Hinglish: *Pehle minus karo (differences), phir divide karo (ratios). Ek bhi test pass nahi hua toh 'neither' - jaise squares aur primes. (iv) dekho: girti hui list bhi GP ho sakti hai, r = 1/2 se har term aadhi ho jaati hai.*

QUESTION 22 | Basic | Terms and membership for 5n-3 (T3/T4)

Q22. For the sequence $t_n = 5n - 3$: (i) find the 10th and 15th terms. (ii) Determine whether 97 and 172 are terms of this sequence.

SOLUTION — step by step

Step 1. (i) $t_{10} = 5 \times 10 - 3 = 47$; $t_{15} = 5 \times 15 - 3 = 72$.

Step 2. (ii) Test 97: 5n - 3 = 97 gives 5n = 100, n = 20 - a natural number, so 97 is the 20th term.

Step 3. Test 172: 5n - 3 = 172 gives 5n = 175, n = 35 - natural, so 172 is the 35th term.

Step 4. Quick sanity check: terms of 5n - 3 end in 2 or 7 (3, 8, 13, 18, ... actually units cycle 3, 8). 97 and 172 both fit the pattern.

» **Why this step?** Membership = solve for n. If n is a natural number, you have also found the position for free.

Answer: (i) 47 and 72 (ii) both are terms: 97 is the 20th, 172 is the 35th.

Hinglish: *Tino kaam ek hi machine se: 5n - 3. 10th? 47. 97 is list ka member hai? n = 20 nikla - haan, 20th term. 172? n = 35 - haan. Solve karo, n whole number hai ya nahi, bas yahi decision criterion hai.*

QUESTION 23 | Basic | Spot the AP (T7)

Q23. Which of the following are arithmetic progressions? Give the common difference when they are: (i) 2, 5, 8, 11 (ii) 1, 4, 9, 16 (iii) 20, 17, 14, 11 (iv) 1, 2, 4, 8 (v) 5, 5, 5, 5

SOLUTION — step by step

Step 1. (i) 5 - 2 = 3, 8 - 5 = 3, 11 - 8 = 3: AP, d = 3.

Step 2. (ii) 3, 5, 7 - differences grow: not an AP (squares).

Step 3. (iii) -3, -3, -3: AP with NEGATIVE difference d = -3 (a decreasing AP).

Step 4. (iv) Ratios are constant (2), not differences (1, 2, 4): GP, not AP.

Step 5. (v) 0, 0, 0: AP with d = 0 - a constant sequence is a legal AP.

» **Why this step?** d can be positive (growing), negative (shrinking) or zero (constant) - all three are genuine APs.

Answer: (i) AP, d = 3 (iii) AP, d = -3 (v) AP, d = 0; (ii) and (iv) are not APs.

Hinglish: *AP ka sirf ek test: consecutive gaps barabar? (iii) mein list ghat rahi hai par gap constant - toh AP hi hai, d = -3. (v) jaise sab 5 - ye bhi AP hai (d = 0). Students yahi do cases miss karte hain.*

QUESTION 24 | Basic | AP terms via a + (n-1)d (T8)

Q24. For the AP 3, 8, 13, 18, ...: (i) state a and d. (ii) Find the 10th and 26th terms. (iii) Write the explicit rule t_n.

SOLUTION — step by step

Step 1. (i) a = 3; d = 8 - 3 = 5 (verify: 13 - 8 = 5 ✓).

Step 2. (ii) $t_{10} = 3 + (10 - 1) \times 5 = 3 + 45 = 48$.

Step 3. $t_{26} = 3 + (26 - 1) \times 5 = 3 + 125 = 128$.

Step 4. (iii) $t_n = 3 + (n - 1) \times 5 = 5n - 2$.

Step 5. Cross-check with the previous question: 5n - 2 with $t_{10} = 48$ and $t_{15} = 72$ matches this AP - same sequence, two descriptions.

» **Why this step?** $t_n = a + (n - 1)d$ works because you climb d exactly (n - 1) times from a. Expanding it gives a linear rule (coefficient of n = d).

Answer: (i) a = 3, d = 5 (ii) $t_{10} = 48$, $t_{26} = 128$ (iii) $t_n = 5n - 2$

Hinglish: *a pehla term, d step size. 10th term = 3 + 9 baar step 5 = 48. Aur $t_n = 5n - 2$ - dekho, coefficient 5 hi d hai aur constant a - d = 3 - 5 = -2. Dono form ek hi kahani.*

QUESTION 25 | Basic | Decreasing AP (T7/T8)

Q25. Consider the AP 11, 7, 3, -1, -5, ... (i) State a and d. (ii) Write the explicit rule t_n. (iii) Find the 10th term.

SOLUTION — step by step

Step 1. (i) a = 11; d = 7 - 11 = -4 (verify: 3 - 7 = -4 ✓). Negative d means the AP decreases.

Step 2. (ii) $t_n = 11 + (n - 1)(-4) = 11 - 4n + 4 = 15 - 4n$.

Step 3. Check: $t_1 = 15 - 4 = 11$ ✓, $t_5 = 15 - 20 = -5$ ✓.

Step 4. (iii) $t_{10} = 15 - 40 = -25$.

» **Why this step?** Substituting d = -4 into the general formula is no different from any other d - the algebra keeps the sign for you.

Answer: (i) a = 11, d = -4 (ii) $t_n = 15 - 4n$ (iii) $t_{10} = -25$

Hinglish: *Girti hui AP - har step -4. Formula mein -4 ghusao aur simplify: 15 - 4n. 10th term = -25, bilkul negative - koi baat nahi, terms negative ho sakti hain, position nahi.*

QUESTION 26 | Basic | Building APs from a and d (T7)

Q26. Write the first four terms of the AP when: (i) a = 5, d = 3; (ii) a = 9, d = -2. In each case state whether the AP increases or decreases.

SOLUTION — step by step

Step 1. (i) Start at 5 and add 3 four times minus one: 5, 8, 11, 14. (Strictly, the 4th term is 5 + 3 × 3 = 14.)

Step 2. Since d = 3 is positive, this AP increases.

Step 3. (ii) Start at 9 and add -2 each time: 9, 7, 5, 3.

Step 4. Since d = -2 is negative, this AP decreases.

» **Why this step?** The AP is completely determined by just two numbers: the start a and the step d - that is the deepest fact about APs.

Answer: (i) 5, 8, 11, 14 - increasing (ii) 9, 7, 5, 3 - decreasing

Hinglish: *AP banane ka recipe: a se shuru, d baar-baar jodo. (ii) mein d negative hai isliye 9 se neeche utarte gaye: 7, 5, 3. Sirf do numbers se poori infinite list control ho jaati hai - amazing hai jab dhyan se socho.*

QUESTION 27 | Basic | Fractional and decimal APs (T7/T8)

Q27. Find the nth term of: (i) the AP 1/2, 5/2, 9/2, 13/2, ... and its 10th term; (ii) the AP 1.5, 3.5, 5.5, 7.5, ... and its 8th term.

SOLUTION — step by step

Step 1. (i) a = 1/2, d = 5/2 - 1/2 = 2 (verify 9/2 - 5/2 = 2 ✓).

Step 2. $t_n = 1/2 + (n - 1) \times 2 = 2n - 3/2$.

Step 3. $t_{10} = 20 - 3/2 = 37/2 = 18.5$.

Step 4. (ii) a = 1.5, d = 3.5 - 1.5 = 2. $t_n = 1.5 + (n - 1) \times 2 = 2n - 0.5$.

Step 5. $t_8 = 16 - 0.5 = 15.5$.

» **Why this step?** Fractions and decimals ride the same formula - a and d are just numbers. Simplify $t_n = a + (n - 1)d$ into 'd times n plus a constant' for speed.

Answer: (i) $t_n = 2n - 3/2$, $t_{10} = 37/2$ (ii) $t_n = 2n - 0.5$, $t_8 = 15.5$

Hinglish: *Fraction/decimal hone se formula nahi badalta. Dono APs ka d = 2 hi nikla - bas shuruaat alag. 2n - 3/2 aur 2n - 0.5: coefficient of n = d, yeh shortcut yaad rakho.*

QUESTION 28 | Basic | Gauss pairing and S20 (T10)

Q28. Use the pairing method to find 1 + 2 + 3 + ... + 10. Then use the formula to find S_{20}.

SOLUTION — step by step

Step 1. Write S = 1 + 2 + ... + 10 and again backwards S = 10 + 9 + ... + 1.

Step 2. Add column-wise: every column gives 11, and there are 10 columns, so 2S = 110.

Step 3. Hence S = 55.

Step 4. Formula route: $S_n = n(n + 1)/2$, so $S_{20} = 20 \times 21/2 = 210$.

» **Why this step?** Pairing first term with last turns a long addition into one multiplication - the idea Āryabhaṭa recorded and Gauss made famous.

Answer: 1 + ... + 10 = 55; $S_{20} = 210$.

Hinglish: *Aage se likho, peechhe se likho, jodo - har column ka total 11, das columns, toh sum aadha = 55. Gauss ko school mein isi trick ne legend bana diya tha. S_{20} ke liye seedha formula: 20 × 21/2 = 210.*

QUESTION 29 | Basic | S50 and S100 (T10)

Q29. Find the sum of the first 50 natural numbers and the sum of the first 100 natural numbers.

SOLUTION — step by step

Step 1. $S_{50} = 50 \times 51/2 = 25 \times 51 = 1275$.

Step 2. $S_{100} = 100 \times 101/2 = 50 \times 101 = 5050$.

Step 3. Both come from one substitution into $S_n = n(n + 1)/2$ - halve the even factor first to keep arithmetic light.

» **Why this step?** n(n + 1)/2 always has one even factor, so halve it mentally before multiplying.

Answer: $S_{50} = 1275$; $S_{100} = 5050$.

Hinglish: *50 × 51/2: pehle 50 aadha karo = 25, phir 25 × 51 = 1275. 100 wale mein 50 × 101 = 5050. Multiplication ko aasan banao - calculator ke bina bhi 5 second mein ho jaata hai.*

QUESTION 30 | Basic | Triangular numbers via the formula (T10)

Q30. Using $t_n = n(n + 1)/2$, find the 10th, 17th and 80th triangular numbers.

SOLUTION — step by step

Step 1. $t_{10} = 10 \times 11/2 = 5 \times 11 = 55$.

Step 2. $t_{17} = 17 \times 18/2 = 17 \times 9 = 153$.

Step 3. $t_{80} = 80 \times 81/2 = 40 \times 81 = 3240$.

» **Why this step?** Triangular numbers ARE the sums S_n - the 17th triangular number is literally 1 + 2 + ... + 17.

Answer: 55, 153, 3240.

Hinglish: *Triangular number = S_n ka fancy naam. 80th triangular number chahiye? 80 se 80 tak likhna pagalpan hota - formula ne 5 second mein 3240 de diya.*

QUESTION 31 | Basic | GP identification with tricky cases (T11)

Q31. Check whether each is a geometric progression and find r when it is: (i) 1, 2, 4, 8, 16, ... (ii) 1, -1, 1, -1, 1, ... (iii) 2, 4, 7, 12, ...

SOLUTION — step by step

Step 1. (i) 2/1 = 2, 4/2 = 2, 8/4 = 2, 16/8 = 2 - all ratios equal 2: GP with r = 2.

Step 2. (ii) -1/1 = -1, 1/-1 = -1, -1/1 = -1 - constant ratio: GP with r = -1. The terms simply alternate 1, -1, 1, -1 forever.

Step 3. (iii) 4/2 = 2 but 7/4 = 1.75 - ratios differ: NOT a GP. (Differences 2, 3, 5 also differ, so it is not an AP either.)

» **Why this step?** A GP needs EVERY consecutive ratio equal - one bad ratio kills it. r = -1 is the smallest legal example of an alternating GP.

Answer: (i) GP, r = 2 (ii) GP, r = -1 (alternating) (iii) not a GP.

Hinglish: *(ii) sabse interesting: 1, -1, 1, -1 bhi GP hai! Ratio -1 se term baari-baari palatti hai. (iii) mein pehla ratio 2 tha, doosra 1.75 - ek hi galat ratio poora GP khatam kar deta hai.*

QUESTION 32 | Basic | Bacteria doubling (T13)

Q32. The number of bacteria in a culture doubles every hour. There are 30 bacteria initially. How many bacteria are present at the end of the 2nd hour, the 4th hour, and the nth hour?

SOLUTION — step by step

Step 1. Start: 30. After each hour the count multiplies by 2 - a GP with a = 30, r = 2.

Step 2. End of 1st hour: 60; end of 2nd hour: $120 = 30 \times 2^2$.

Step 3. End of 4th hour: $30 \times 2^4 = 30 \times 16 = 480$.

Step 4. End of nth hour: 30×2^n.

» **Why this step?** After n hours the culture has doubled n times, so the factor is 2^n - count the doublings, that is the exponent.

Answer: 2nd hour: 120; 4th hour: 480; nth hour: 30×2^n.

Hinglish: *Double hote raha = GP with r = 2. n ghante mein n baar double = 2^n ka factor. $30 \times 2^4 = 480$. Aise growth questions mein multiply count karo - hours hi exponent ban jaate hain.*

Level 2 · Core Drills: book-level practice:

QUESTION 33 | Moderate | Which term is 607 (T4/T8)

Q33. Which term of the sequence $t_n = 5n - 3$ is 607?

SOLUTION — step by step

Step 1. Set $5n - 3 = 607$.

Step 2. $5n = 610$.

Step 3. $n = 122$.

Step 4. n is a natural number, so 607 is the 122nd term of the sequence.

» **Why this step?** 'Which term' questions are linear equations in disguise: solve, then confirm n is a counting number.

Answer: The 122nd term.

Hinglish: *Seedha equation: 5n - 3 = 607, n = 122. Whole number aaya toh answer pakka. Fraction aata toh answer hota 'not a term' - yahan toh clean 122 nikla.*

QUESTION 34 | Moderate | Three-term recursion (T5)

Q34. Let $T_1 = 1$, $T_2 = 2$, $T_3 = 4$, and $T_n = T_{n-1} + T_{n-2} + T_{n-3}$ for n at least 4. Find T_4, T_5, T_6, T_7 and T_8.

SOLUTION — step by step

Step 1. $T_4 = T_3 + T_2 + T_1 = 4 + 2 + 1 = 7$.

Step 2. $T_5 = T_4 + T_3 + T_2 = 7 + 4 + 2 = 13$.

Step 3. $T_6 = T_5 + T_4 + T_3 = 13 + 7 + 4 = 24$.

Step 4. $T_7 = T_6 + T_5 + T_4 = 24 + 13 + 7 = 44$.

Step 5. $T_8 = T_7 + T_6 + T_5 = 44 + 24 + 13 = 81$.

» **Why this step?** This 'tribonacci' rule needs THREE starting seeds and adds the last three terms - the window simply slides forward one step at a time.

Answer: 7, 13, 24, 44, 81.

Hinglish: *Teen pichhle terms ka sum - tribonacci! Window slide karo: 1, 2, 4 se 7; phir 2, 4, 7 se 13... Har baar purani sabse purani term hatao, nayi jodo. Sliding window ka khel hai.*

QUESTION 35 | Moderate | Recursive AP and membership (T5/T4)

Q35. A sequence is given by the recursive rule $t_1 = -5$, $t_{n+1} = t_n + 3$ for n at least 1. Find the first five terms. Is 52 a term of this sequence? If so, which one?

SOLUTION — step by step

Step 1. $t_1 = -5$; $t_2 = -5 + 3 = -2$; $t_3 = -2 + 3 = 1$; $t_4 = 1 + 3 = 4$; $t_5 = 4 + 3 = 7$.

Step 2. First five terms: -5, -2, 1, 4, 7 - an AP with a = -5, d = 3.

Step 3. Explicit rule: $t_n = -5 + (n - 1) \times 3 = 3n - 8$.

Step 4. Test 52: 3n - 8 = 52 gives 3n = 60, n = 20 - a natural number.

Step 5. So 52 is a term: exactly the 20th.

» **Why this step?** The recursion '+3 each time' is secretly an AP - converting the recursive language into $t_n = 3n - 8$ makes the membership test one line.

Answer: -5, -2, 1, 4, 7; yes - 52 is the 20th term.

Hinglish: *Recursive +3 dikha toh turant AP samjho: a = -5, d = 3. Formula 3n - 8 banao, 52 ke liye n = 20 - haan, 20th term. Recursive se explicit mein translate karna hi asli skill hai.*

QUESTION 36 | Moderate | Membership in 5n-2 (T4)

Q36. For the sequence $s_n = 5n - 2$, determine whether 308 is a term. Do the same for 461. In each case give a reason tied to the value of n you find.

SOLUTION — step by step

Step 1. Test 308: 5n - 2 = 308 gives 5n = 310, n = 62.

Step 2. n = 62 is a natural number, so 308 is a term - the 62nd.

Step 3. Test 461: 5n - 2 = 461 gives 5n = 463, n = 463/5 = 92.6.

Step 4. n = 92.6 is not a natural number (positions must be whole), so 461 is NOT a term.

» **Why this step?** The test is airtight: every term of 5n - 2 equals 5 × (whole number) - 2, so it must end in 2 or 7. 461 ends in 1 - doomed from the start.

Answer: 308 is a term (62nd); 461 is not a term (n = 92.6 is not natural).

Hinglish: *308 nikal gaya (n = 62). 461 atka: n = 92.6 - aadha term exist nahi karta! Shortcut: 5n - 2 ke terms hamesha 2 ya 7 par khatam hote hain; 461 toh 1 par khatam hai - reject before solving.*

QUESTION 37 | Moderate | Position in the odd numbers (T4)

Q37. The odd number 137 appears in the sequence 1, 3, 5, 7, ... Find the position it occupies using the explicit rule $u_n = 2n - 1$.

SOLUTION — step by step

Step 1. Set $u_n = 137$: $2n - 1 = 137$.

Step 2. $2n = 138$.

Step 3. $n = 69$.

Step 4. So 137 is the 69th odd number. (Check: the 69th odd number is $2 \times 69 - 1 = 137$ ✓.)

» **Why this step?** Every odd number 2k - 1 sits at position k - the rule makes positions transparent.

Answer: The 69th position.

Hinglish: *2n - 1 = 137, n = 69. Matlab 137 tak pahunchne ke liye 68 baar +2 karna pada. Odd number ka position uske value se ek second mein: (value + 1)/2.*

QUESTION 38 | Moderate | Verify AP, rule and plot (T7/T8/T14)

Q38. (i) Verify that 2, 5, 8, 11, ... is an AP and write its nth term. (ii) If you plot the pairs (1, 2), (2, 5), (3, 8), (4, 11), what shape do you expect and why?

SOLUTION — step by step

Step 1. (i) Differences: 5 - 2 = 3, 8 - 5 = 3, 11 - 8 = 3 - constant, so it is an AP with a = 2, d = 3.

Step 2. $t_n = 2 + (n - 1) \times 3 = 3n - 1$.

Step 3. (ii) The plotted points must lie on ONE STRAIGHT LINE.

Step 4. Reason: $t_n = 3n - 1$ is exactly the line $y = 3x - 1$ evaluated at whole x - a linear rule graphs as a line, and equal x-steps produce equal y-steps of 3.

» **Why this step?** AP and straight line are the same fact in two dresses: constant slope = constant difference.

Answer: (i) AP with a = 2, d = 3; $t_n = 3n - 1$ (ii) all points on the straight line $y = 3x - 1$.

Hinglish: *Constant difference = constant slope. Points plot karo toh seedhi line milegi y = 3x - 1. Isliye AP ko 'linear pattern' kehte hain - graph khud formula dikha deta hai.*

QUESTION 39 | Moderate | Negative-start AP rule (T7/T8)

Q39. For the sequence -5, -1, 3, 7, ...: (i) verify it is an AP and write t_n. (ii) Find the 50th term.

SOLUTION — step by step

Step 1. (i) Differences: -1 - (-5) = 4, 3 - (-1) = 4, 7 - 3 = 4 - constant d = 4. So yes, an AP with a = -5.

Step 2. $t_n = -5 + (n - 1) \times 4 = 4n - 9$.

Step 3. Check: $t_1 = 4 - 9 = -5$ ✓; $t_2 = 8 - 9 = -1$ ✓.

Step 4. (ii) $t_{50} = 4 \times 50 - 9 = 200 - 9 = 191$.

» **Why this step?** Subtracting a negative needs care: -1 - (-5) = -1 + 5 = 4. Most sign errors in APs happen in exactly this step.

Answer: (i) AP with d = 4; $t_n = 4n - 9$ (ii) $t_{50} = 191$.

Hinglish: *Minus se minus ka subtraction dhyan se: -1 - (-5) = +4. Ye ek step galti hoti hai sabse zyada. Baaki sab routine: 4n - 9, aur 50th = 191.*

QUESTION 40 | Moderate | Which term is -81; is 0 a term (T4/T8)

Q40. Consider the AP 21, 18, 15, ... (i) Which term of this AP is -81? (ii) Is 0 a term of this AP? Give reasons.

SOLUTION — step by step

Step 1. a = 21, d = 18 - 21 = -3.

Step 2. (i) Set a + (n - 1)d = -81: 21 - 3(n - 1) = -81.

Step 3. -3(n - 1) = -102, so n - 1 = 34, n = 35. The 35th term is -81.

Step 4. (ii) Test 0: 21 - 3(n - 1) = 0 gives 3(n - 1) = 21, n - 1 = 7, n = 8.

Step 5. n = 8 is a natural number, so 0 IS a term - list it: 21, 18, 15, 12, 9, 6, 3, 0. Yes, the 8th term is exactly 0.

» **Why this step?** Students 'feel' a descending list can never hit 0 - but the algebra says otherwise. Trust the equation, not the feeling.

Answer: (i) the 35th term (ii) yes - 0 is the 8th term (21 - 3 × 7 = 0).

Hinglish: *-81 35th term par mila. Aur twist: 0 bhi term hai - 8th! List likh ke dekho: 21, 18, 15, 12, 9, 6, 3, 0. Ghat'ti hui AP zero se guzar sakti hai. Equation ko haq do, intuition ko nahi.*

QUESTION 41 | Moderate | Explicit plus recursive rule of an AP (T7/T8)

Q41. For the AP 11, 8, 5, 2, ...: (i) find the nth term. (ii) Write its recursive rule. (iii) Find the 15th term.

SOLUTION — step by step

Step 1. a = 11, d = 8 - 11 = -3 (verify 5 - 8 = -3 ✓).

Step 2. (i) $t_n = 11 + (n - 1)(-3) = 14 - 3n$.

Step 3. (ii) Recursive rule: $t_1 = 11$, $t_n = t_{n-1} - 3$ for n at least 2 - 'each term is 3 less than the previous one'.

Step 4. (iii) $t_{15} = 14 - 45 = -31$.

» **Why this step?** Every AP has two complete descriptions: explicit (14 - 3n) and recursive (start 11, subtract 3). Both generate the same list.

Answer: (i) $t_n = 14 - 3n$ (ii) $t_1 = 11, t_n = t_{n-1} - 3$ (iii) -31

Hinglish: *Ek hi AP, do zubaan: explicit 14 - 3n aur recursive '11 se shuru, har baar -3'. Question jo maange woh do. 15th term = -31 - negative aana natural hai kyunki d bhi negative hai.*

QUESTION 42 | Moderate | AP from two conditions (T9)

Q42. An AP consists of 50 terms. Its 3rd term is 12 and its last (50th) term is 106. Find the 29th term.

SOLUTION — step by step

Step 1. Let the AP have first term a and common difference d.

Step 2. 3rd term: a + 2d = 12.

Step 3. 50th term: a + 49d = 106.

Step 4. Subtract: 47d = 94, so d = 2. Then a = 12 - 2d = 12 - 4 = 8.

Step 5. $t_{29} = 8 + 28 \times 2 = 64$.

» **Why this step?** Two facts about two different terms = two linear equations in a and d. Subtracting kills a and exposes d instantly.

Answer: d = 2, a = 8; $t_{29} = 64$.

Hinglish: *Do equations, do unknowns - classic combo. 47d = 94 se d = 2, wapas daal ke a = 8. 29th term = 64. Ye pattern itna common hai ki exam mein ye question 'free marks' hai agar method yaad ho.*

QUESTION 43 | Moderate | Salary increment word problem (T8/T13)

Q43. Harish starts work at an annual salary of ₹5,00,000 and receives an increment of ₹20,000 each year. After how many years does his income reach ₹7,00,000?

SOLUTION — step by step

Step 1. Salaries form an AP: 5,00,000; 5,20,000; 5,40,000; ... with a = 5,00,000 and d = 20,000.

Step 2. nth-year salary: $t_n = 5,00,000 + (n - 1) \times 20,000$.

Step 3. Set $t_n = 7,00,000$: $(n - 1) \times 20,000 = 2,00,000$.

Step 4. $n - 1 = 10$, so $n = 11$.

Step 5. Check: after 1 year (n = 1) he is on the starting salary; the 11th year's salary is 7,00,000. So his income reaches ₹7,00,000 in the 11th year, i.e. after 10 increments.

» **Why this step?** Count carefully whether n counts years-served or the year index - the (n - 1) in the formula encodes exactly that choice.

Answer: In the 11th year (after 10 annual increments) his salary reaches ₹7,00,000.

Hinglish: *Increment wala AP: 5L se shuru, har saal 20K upar. 2L ka total uplift chahiye = 10 increments = 11th year. Off-by-one yahan sabse bada dhoka hai - 'kitne saal baad' vs 'kaunsa saal' - dono padho dhyan se.*

QUESTION 44 | Moderate | Marbles in rows (T10)

Q44. A child arranges marbles in rows: the first row has 1 marble, the second has 2, the third has 3, and so on up to 25 rows. How many marbles does the child use in all?

SOLUTION — step by step

Step 1. Total = $1 + 2 + 3 + ... + 25 = S_{25}$.

Step 2. $S_{25} = 25 \times 26/2$.

Step 3. $= 25 \times 13 = 325$.

Step 4. So the child uses 325 marbles.

» **Why this step?** The row pattern IS the triangular numbers - the sum formula n(n + 1)/2 finishes it in one line.

Answer: 325 marbles.

Hinglish: *Rows 1, 2, 3, ... 25 - ye toh direct S_{25} hai. 25 × 26/2 = 325. Pattern pehchano, formula lagao, done. Varna 25 numbers add karne mein hi 2 minute jayenge.*

QUESTION 45 | Moderate | Two-digit multiples of 3: count and sum (T8/T10)

Q45. How many two-digit numbers are divisible by 3? What is the sum of all these two-digit numbers?

SOLUTION — step by step

Step 1. Two-digit multiples of 3: smallest 12, largest 99. They form the AP 12, 15, 18, ..., 99 with d = 3.

Step 2. Count: 12 + (n - 1) × 3 = 99 gives (n - 1) × 3 = 87, n - 1 = 29, n = 30. So there are 30 such numbers.

Step 3. Sum by the average idea: sum = (first + last)/2 × count = (12 + 99)/2 × 30.

Step 4. = (111/2) × 30 = 111 × 15 = 1665.

» **Why this step?** Any 'multiples of k between...' question is an AP in disguise: find first and last, count by the nth-term equation, sum by average × count.

Answer: 30 numbers; their sum is 1665.

Hinglish: *12 se 99 tak 3 ke multiples = AP. Count: (99-12)/3 + 1 = 30. Sum ka Āryabhaṭa trick: average × count = 55.5 × 30 = 1665. Pehla-akhirka jodo, aadha karo, count se guna karo - formula rattne ki zaroorat nahi.*

QUESTION 46 | Moderate | Range sum 25 to 58 (T10)

Q46. Find the sum 25 + 26 + 27 + ... + 58 without adding term by term.

SOLUTION — step by step

Step 1. Cut off the head: 25 + ... + 58 = (1 + 2 + ... + 58) - (1 + 2 + ... + 24) = S_{58} - S_{24}.

Step 2. S_{58} = 58 × 59/2 = 29 × 59 = 1711.

Step 3. S_{24} = 24 × 25/2 = 12 × 25 = 300.

Step 4. Sum = 1711 - 300 = 1411.

Step 5. Alternative check: 34 terms, average (25 + 58)/2 = 41.5, and 34 × 41.5 = 1411 ✓.

» **Why this step?** Range sum = bigger staircase minus smaller staircase. The average × count route double-checks it in one line.

Answer: 1411.

Hinglish: *Beech ka sum = bada S minus chhota S. 1711 - 300 = 1411. Aur verification: 34 terms × average 41.5 = 1411. Dono tareeke match - ab pack up.*

QUESTION 47 | Moderate | Range sum 40 to 70 (T10)

Q47. Compute 40 + 41 + ... + 70 two different ways and confirm both give the same answer.

SOLUTION — step by step

Step 1. Way 1 (S difference): $S_{70} - S_{39} = 70 \times 71/2 - 39 \times 40/2 = 2485 - 780 = 1705$.

Step 2. Way 2 (average × count): count = 70 - 40 + 1 = 31 terms; average = (40 + 70)/2 = 55.

Step 3. Sum = 55 × 31 = 1705.

Step 4. Both roads land on 1705 - the two formulas are algebraically the same thing.

» **Why this step?** Counting an inclusive range is (last - first) + 1 - forgetting the +1 is the classic off-by-one error here.

Answer: 1705 by both methods.

Hinglish: *Count mein +1 mat bhoolo: 70 - 40 + 1 = 31 terms. Average 55. 55 × 31 = 1705. Dono methods ka match mil gaya - ab bina dare answer likho.*

QUESTION 48 | Moderate | Stacked logs (T10)

Q48. A stack of logs has 20 logs in the bottom row, 19 in the row above, and so on, with 1 log at the top. How many logs are in the stack?

SOLUTION — step by step

Step 1. Rows hold 20, 19, 18, ..., 1 logs - the numbers 1 to 20 in reverse order (order does not affect a sum).

Step 2. Total = $S_{20} = 20 \times 21/2$.

Step 3. = 10 × 21 = 210.

» **Why this step?** A decreasing list sums the same as its reverse - recognise 1 + 2 + ... + 20 even when it is written backwards.

Answer: 210 logs.

Hinglish: *Ulta likha hua 1 se 20 tak ka sum hai - jodne par farak nahi padta. 210 logs. Pattern ulta aaye toh palto aur S_n lagao.*

QUESTION 49 | Moderate | GP 5, 25, 125 - far terms (T12)

Q49. For the GP 5, 25, 125, ...: (i) find the 10th term. (ii) Write the nth term. (iii) Find the 20th term.

SOLUTION — step by step

Step 1. a = 5; r = 25/5 = 5 (verify 125/25 = 5 ✓).

Step 2. (i) $t_{10} = 5 \times 5^9 = 5^{10} = 9{,}765{,}625$.

Step 3. (ii) $t_n = 5 \times 5^{n-1} = 5^n$ - the powers fold neatly because a = r.

Step 4. (iii) $t_{20} = 5^{20} = 95{,}367{,}431{,}640{,}625$ (about 9.5×10^{13}).

» **Why this step?** When a equals r, $t_n = r^n$ - the two factors merge. Powers of 5 grow brutally: 20 steps and you are in the tens of trillions.

Answer: (i) 9,765,625 (ii) $t_n = 5^n$ (iii) 95,367,431,640,625.

Hinglish: *5, 25, 125 - ye 5 ke powers hain: 5^1, 5^2, 5^3... Toh nth term 5^n. 20th term mein 14 digits! GP ka exponential uddan yahi hai - 20 steps mein trillions.*

QUESTION 50 | Moderate | Which term of a GP is 4374 (T12)

Q50. Which term of the GP 2, 6, 18, ... is 4374? Write the explicit formula and the recursive formula for the nth term of this GP.

SOLUTION — step by step

Step 1. a = 2; r = 6/2 = 3 (verify 18/6 = 3 ✓).

Step 2. Explicit formula: $t_n = 2 \times 3^{n-1}$.

Step 3. Set $2 \times 3^{n-1} = 4374$, so $3^{n-1} = 2187$.

Step 4. $2187 = 3^7$ (since $3^7 = 2187$), so n - 1 = 7 and n = 8.

Step 5. Recursive formula: $t_1 = 2$, $t_n = 3 \cdot t_{n-1}$ for n at least 2.

Step 6. 4374 is the 8th term. Check: 2, 6, 18, 54, 162, 486, 1458, 4374 ✓.

» **Why this step?** 'Which term of a GP' = solve $ar^{n-1} = k$ by matching powers. Knowing small powers of 3 (up to $3^7 = 2187$) makes this instant.

Answer: The 8th term; explicit $t_n = 2 \times 3^{n-1}$; recursive $t_1 = 2$, $t_n = 3t_{n-1}$.

Hinglish: *$4374 = 2 \times 2187 = 2 \times 3^7$. Toh n - 1 = 7, matlab 8th term. Powers of 3 yaad rakho: 3, 9, 27, 81, 243, 729, 2187 - GP questions half-time mein hote hain. Recursive form: har term ko 3 se multiply.*

QUESTION 51 | Moderate | GP with fractional ratio (T11/T12)

Q51. Check whether 5, 15/4, 45/16, 135/64, ... is a geometric progression. If it is, find r and the nth term.

SOLUTION — step by step

Step 1. $t_2/t_1 = (15/4)/5 = 15/4 \times 1/5 = 3/4$.

Step 2. $t_3/t_2 = (45/16)/(15/4) = 45/16 \times 4/15 = 3/4$.

Step 3. $t_4/t_3 = (135/64)/(45/16) = 135/64 \times 16/45 = 3/4$.

Step 4. All ratios equal 3/4 - it is a GP with a = 5, r = 3/4.

Step 5. $t_n = 5 \times (3/4)^{n-1}$.

» **Why this step?** Dividing by a fraction = multiplying by its reciprocal. A fractional ratio below 1 makes the GP shrink towards 0 - decay, not growth.

Answer: Yes - a GP with a = 5, r = 3/4; $t_n = 5 \times (3/4)^{n-1}$.

Hinglish: *Fraction divide karo toh ulta multiply hota hai: (15/4) ÷ 5 = 15/4 × 1/5 = 3/4. Teeno ratios 3/4 - GP pakka. r < 1 hai toh terms girti jayengi - 5, 3.75, 2.8125... towards zero.*

QUESTION 52 | Moderate | GP verification drill (T11/T12)

Q52. Check whether each sequence is a geometric progression and find its nth term when it is: (i) 2, 10, 50, 250, ... (ii) 4, 8/3, 16/9, 32/27, ...

SOLUTION — step by step

Step 1. (i) Ratios: 10/2 = 5, 50/10 = 5, 250/50 = 5 - GP with a = 2, r = 5.

Step 2. $t_n = 2 \times 5^{n-1}$.

Step 3. (ii) Ratios: (8/3)/4 = 8/12 = 2/3; (16/9)/(8/3) = 16/9 × 3/8 = 2/3; (32/27)/(16/9) = 32/27 × 9/16 = 2/3 - GP with a = 4, r = 2/3.

Step 4. $t_n = 4 \times (2/3)^{n-1}$.

» **Why this step?** The ratio test is three divisions - do all of them. A shrinking ratio like 2/3 is as legitimate as a growing one like 5.

Answer: (i) GP, $t_n = 2 \times 5^{n-1}$ (ii) GP, $t_n = 4 \times (2/3)^{n-1}$.

Hinglish: *Dono GP hain - ek udti hui (r = 5), ek girti hui (r = 2/3). Fraction ratio mein reciprocal multiply karo aur simplify. nth term ka format same: $a \times r^{n-1}$.*

QUESTION 53 | Moderate | GP with negative ratio (T11/T12)

Q53. The sequence 1/2, -1/4, 1/8, -1/16, ... alternates in sign. (i) Show it is a GP and find r. (ii) Find the 6th and 10th terms.

SOLUTION — step by step

Step 1. (i) Ratios: (-1/4)/(1/2) = -1/2; (1/8)/(-1/4) = -1/2; (-1/16)/(1/8) = -1/2 - GP with a = 1/2, r = -1/2.

Step 2. $t_n = (1/2) \times (-1/2)^{n-1}$.

Step 3. (ii) $t_6 = (1/2) \times (-1/2)^5 = (1/2) \times (-1/32) = -1/64$.

Step 4. $t_{10} = (1/2) \times (-1/2)^9 = (1/2) \times (-1/512) = -1/1024$.

» **Why this step?** A negative ratio makes terms alternate: odd n gives positive terms here, even n negative (check: t_6, t_{10} are both negative - both even positions).

Answer: (i) GP with a = 1/2, r = -1/2; $t_n = (1/2)(-1/2)^{n-1}$ (ii) $t_6 = -1/64$, $t_{10} = -1/1024$.

Hinglish: *r = -1/2 se signs baari-baari palatte hain. Even position par minus, odd par plus (is sequence mein). Magnitude har baar aadhi - 1/2, 1/4, 1/8... Decay plus oscillation - dono ek saath.*

QUESTION 54 | Moderate | Recursive form of a GP (T5/T12)

Q54. The GP 3, 30, 300, 3000, ... is generated by the explicit rule $t_n = 3 \times 10^{n-1}$. Find a recursive rule for the same sequence, and state t_7.

SOLUTION — step by step

Step 1. Each term is 10 times the one before it: 30 = 10 × 3, 300 = 10 × 30, 3000 = 10 × 300.

Step 2. Recursive rule: $t_1 = 3$, $t_n = 10 \cdot t_{n-1}$ for n at least 2.

Step 3. $t_7 = 3 \times 10^6 = 3{,}000{,}000$.

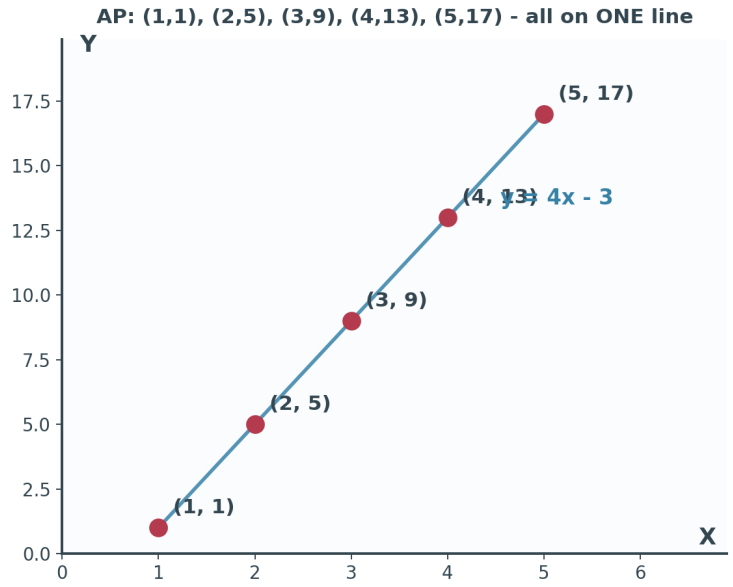
» **Why this step?** Explicit and recursive rules are translations of each other: 'multiply by r every step' (recursive) is 'multiply by r^{n-1} in total' (explicit).

Answer: $t_1 = 3$, $t_n = 10\ t_{n-1}$; $t_7 = 3{,}000{,}000$.

Hinglish: *10 se multiply hota rahega - recursive yahi bolta hai. 7th term: 3 followed by 6 zeros = 30 lakh. Explicit aur recursive ek hi sequence ki do tasveerein hain - jo question maange woh dena.*

QUESTION 55 | Moderate | AP plotted as a line (T14)

Q55. The figure shows the points (1, 1), (2, 5), (3, 9), (4, 13), (5, 17) plotted from the growing-square counts 1, 5, 9, 13, 17. (i) Verify from the graph that the points are collinear. (ii) Find the equation of the line. (iii) Why do AP points always behave like this?



SOLUTION — step by step

Step 1. (i) From point to point: x-step 1, y-step +4 every time (1 to 5 to 9 to 13 to 17). Equal steps in equal x-intervals = constant slope 4 - the points lie on one straight line.

Step 2. (ii) Slope 4 through (1, 1): $y - 1 = 4(x - 1)$, so $y = 4x - 3$. (This is exactly $t_n = 4n - 3$ with $x = n$.)

Step 3. Check (3, 9): $4 \times 3 - 3 = 9$ ✓; check (5, 17): $20 - 3 = 17$ ✓.

Step 4. (iii) An AP adds the same d every step, so its $t_n = a + (n - 1)d$ is linear in n with slope d - every AP plots onto a straight line of slope d.

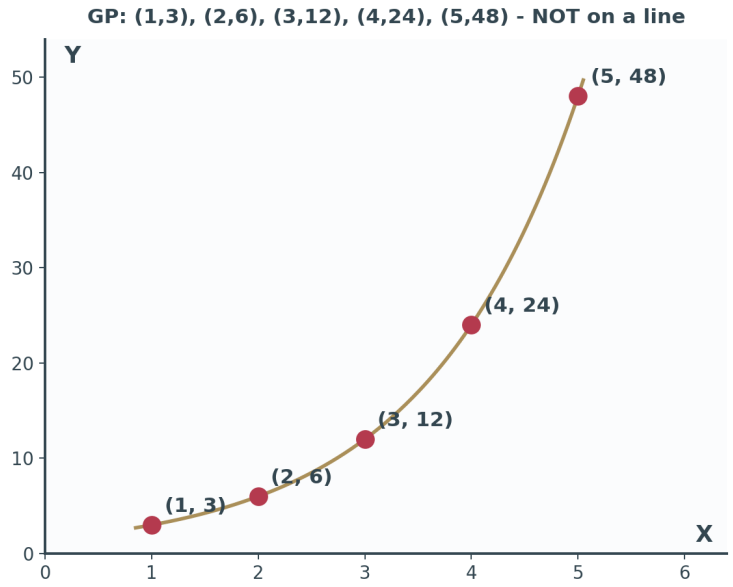
» **Why this step?** The graph of an AP is the line $y = dx + (a - d)$: slope = common difference, and the y-intercept secretly stores a - d.

Answer: (i) collinear with constant slope 4 (ii) y = 4x - 3 (iii) AP growth is linear: same d each step gives one straight line.

Hinglish: *Graph mein har step +4 - constant slope, toh line hi banegi. Equation y = 4x - 3 = wahi 4n - 3. AP ka graph hamesha line hota hai jiska slope d hota hai. Ye connection coordinate geometry ke saath is chapter ka bonus hai.*

QUESTION 56 | Moderate | GP plotted as a curve (T14)

Q56. The figure shows the GP points (1, 3), (2, 6), (3, 12), (4, 24), (5, 48). (i) Do they lie on a straight line? Justify using slopes between consecutive points. (ii) What kind of curve behaviour is this, and which GP feature causes it?



SOLUTION — step by step

Step 1. (i) Slopes: (6 - 3)/1 = 3; (12 - 6)/1 = 6; (24 - 12)/1 = 12; (48 - 24)/1 = 24. The slopes keep doubling - NOT equal, so the points do not lie on one line.

Step 2. (ii) Equal ratios (r = 2) mean the y-values double for each unit x-step: 3, 6, 12, 24, 48 - the curve bends ever more steeply upward.

Step 3. This is exponential-style growth: multiplication by a fixed r > 1 produces curvature, never a line.

Step 4. Contrast: an AP's equal ADDITIONS give equal slopes (a line). A GP's equal MULTIPLIERS give growing slopes (a curve).

» **Why this step?** Constant slope means line (AP); constant ratio means curve (GP). The graph type is a fingerprint of the sequence type.

Answer: (i) No - consecutive slopes 3, 6, 12, 24 are unequal (ii) exponential-style upward curve, caused by the constant multiplier r = 2.

Hinglish: *Slopes 3, 6, 12, 24 - har baar double, toh line nahi ban sakti. GP ka graph curve hota hai jo tezi se ooncha hota jaata hai. Exam mein 'line ya curve?' pooche toh: differences constant = line, ratios constant = curve.*

QUESTION 57 | Moderate | GP with irrational ratio (T12)

Q57. Consider the sequence 2, $2\sqrt{2}$, 4, $4\sqrt{2}$, ... Which term of this sequence is 128?

SOLUTION — step by step

Step 1. Ratios: $2\sqrt{2}/2 = \sqrt{2}$; $4/(2\sqrt{2}) = 2/\sqrt{2} = \sqrt{2}$ - a GP with $a = 2$, $r = \sqrt{2}$.

Step 2. $t_n = 2 \times (\sqrt{2})^{n-1} = 2 \times 2^{(n-1)/2} = 2^{(n+1)/2}$.

Step 3. Set $2^{(n+1)/2} = 128 = 2^7$.

Step 4. $(n + 1)/2 = 7$, so $n + 1 = 14$ and $n = 13$.

Step 5. 128 is the 13th term. (Check by stepping: 2, $2\sqrt{2}$, 4, $4\sqrt{2}$, 8, $8\sqrt{2}$, 16, $16\sqrt{2}$, 32, $32\sqrt{2}$, 64, $64\sqrt{2}$, 128 ✓)

» **Why this step?** Convert every factor to base 2: $(\sqrt{2})^{n-1} = 2^{(n-1)/2}$. Powers of a single base make 'which term' questions mechanical.

Answer: The 13th term.

Hinglish: *$\sqrt{2}$ ko $2^{1/2}$ likho, sab 2 ki powers ban jaati hain. $128 = 2^7$, toh $(n+1)/2 = 7$, $n = 13$. Irrational ratio se mat daro - base convert karo, baaki same game.*

QUESTION 58 | Moderate | Hidden pattern in a recursion (T5/T3)

Q58. A sequence is given by the recursive rule $t_1 = 2$, $t_{n+1} = 3t_n - 2$ for n at least 1. Which term of this sequence is 730? Can you spot a simpler explicit rule hiding in the terms?

SOLUTION — step by step

Step 1. $t_1 = 2$.

Step 2. $t_2 = 3 \times 2 - 2 = 4$; $t_3 = 3 \times 4 - 2 = 10$.

Step 3. $t_4 = 3 \times 10 - 2 = 28$; $t_5 = 3 \times 28 - 2 = 82$; $t_6 = 3 \times 82 - 2 = 244$.

Step 4. $t_7 = 3 \times 244 - 2 = 730$. So 730 is the 7th term.

Step 5. Hidden pattern: add 1 to every term - 3, 5, 11, 29, 83, 245, $731 = 3^1, 3^2, 3^3, \ldots$ So $t_n = 3^{n-1} + 1$. Check: $t_7 = 3^6 + 1 = 729 + 1 = 730$ ✓.

» **Why this step?** When a recursion looks ugly, list a few terms and hunt for a familiar neighbour - here each term is 'one more than a power of 3'.

Answer: The 7th term; hidden explicit rule $t_n = 3^{n-1} + 1$.

Hinglish: *730 7th term par mila. Aur asli jackpot: har term mein +1 karo toh 3 ki powers bachi nikalti hain! $t_n = 3^{n-1} + 1$. Recursive jungle mein explicit sadak dhoondhna hi asli math hai.*

QUESTION 59 | Moderate | Library word problem (T8)

Q59. A library starts with 120 books and adds 5 new books every month. (i) Write the book count as a sequence and give its nth term (n = months elapsed). (ii) How many books after 12 months? (iii) When does the library reach 200 books?

SOLUTION — step by step

Step 1. (i) Counts: 120 (start), 125, 130, 135, ... - an AP with a = 120, d = 5.

Step 2. With n = months elapsed, count = 120 + 5n (the first +5 arrives at n = 1).

Step 3. (ii) After 12 months: 120 + 5 × 12 = 180 books.

Step 4. (iii) Solve 120 + 5n = 200: 5n = 80, n = 16. The library reaches 200 books after 16 months.

» **Why this step?** Choose the meaning of n first (months elapsed vs month number) and state it - the constant in the formula depends on that choice.

Answer: (i) 125, 130, 135, ...; count = 120 + 5n (ii) 180 books (iii) after 16 months.

Hinglish: *Yahan n = mahine guzar chuke (month number nahi) - isliye formula 120 + 5n bana, 120 + (n-1)5 nahi. 12 mahine baad 180. 200 books kab? n = 16. n ki definition pehle fix karo, phir formula aadha apne aap ban jaata hai.*

QUESTION 60 | Moderate | Weekly savings AP (T8)

Q60. Rekha saves ₹50 in the first week and increases her weekly saving by ₹15 every week. (i) Write her weekly savings as a sequence with an nth term (week 1, week 2, ...). (ii) In which week does her weekly saving first reach ₹500?

SOLUTION — step by step

Step 1. (i) Savings: 50, 65, 80, 95, ... - an AP with a = 50, d = 15.

Step 2. Week-n saving: $t_n = 50 + (n - 1) \times 15 = 15n + 35$.

Step 3. (ii) Set 15n + 35 = 500: 15n = 465, so n = 31.

Step 4. n = 31 is a natural number - her weekly saving hits exactly ₹500 in week 31.

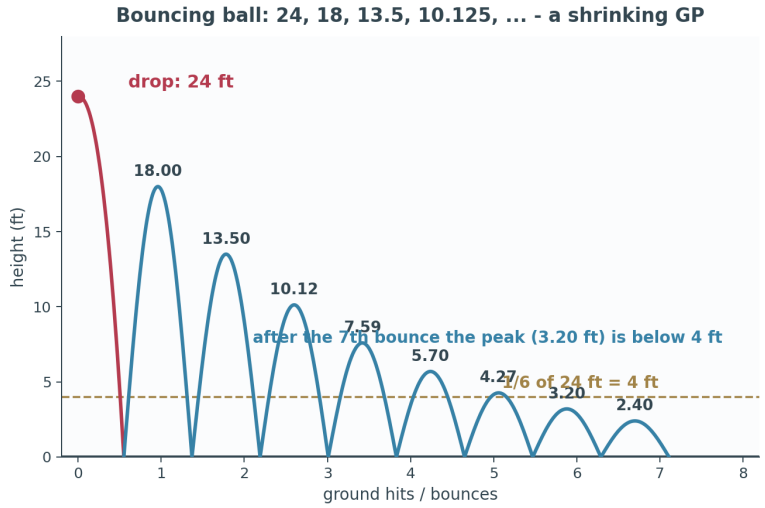
» **Why this step?** When week numbering starts at 1, keep (n - 1) in the formula; expanding to 15n + 35 makes the equation cleaner.

Answer: (i) 50, 65, 80, ...; $t_n = 15n + 35$ (ii) week 31.

Hinglish: *50, 65, 80... classic AP. 15n + 35 = 500 se n = 31 - exactly 31st week mein ₹500. 'First reach' poochha hai toh whole number aana hi chahiye; agar fraction aata toh next whole week lena.*

QUESTION 61 | Moderate | Bouncing ball heights (T13)

Q61. A ball is dropped from a height of 24 feet. After hitting the ground it bounces up to 3/4 of its previous height each time. (i) Write the heights reached after each of the first five bounces. (ii) The heights form a GP - state a and r. (iii) Show that after the 7th bounce the ball stays below 1/6 of its original dropping height.



SOLUTION — step by step

Step 1. (i) Bounce 1: 24 × 3/4 = 18 ft. Bounce 2: 18 × 3/4 = 13.5 ft. Bounce 3: 10.125 ft. Bounce 4: 7.59375 ft. Bounce 5: 5.6953125 ft.

Step 2. (ii) The bounce heights 18, 13.5, 10.125, ... form a GP with first term a = 18 and ratio r = 3/4 (each height is 3/4 of the one before).

Step 3. (iii) 1/6 of 24 ft = 4 ft. Continuing: bounce 6 = 5.6953125 × 3/4 = 4.271484375 ft (still above 4), bounce 7 = 4.271484375 × 3/4 = 3.20361328125 ft.

Step 4. 3.2036... ft < 4 ft, so from the 7th bounce onward the ball never rises above 1/6 of the original height.

Step 5. Formula route: bounce-n height = $18 \times (3/4)^{n-1}$; set it below 4 and count the bounces.

» **Why this step?** The bounce heights START at bounce 1 with value 24 × 3/4 - the drop height 24 is the 'zeroth' term, not the first.

Answer: (i) 18, 13.5, 10.125, 7.59375, 5.6953 ft (ii) a = 18, r = 3/4 (iii) 7th-bounce height 3.2036 ft is below 4 ft = 1/6 of 24 ft.

Hinglish: *Pehla bounce 24 ka 3/4 = 18 ft. Har baar 3/4 - GP with r = 3/4. 24 ka 1/6 = 4 ft. 6th bounce 4.27 (abhi bhi upar), 7th bounce 3.20 - ab 4 ke neeche. Ball ki story bhi GP hai, bas shrinking wali.*

QUESTION 62 | Moderate | Bouncing ball total distance (T13)

Q62. A ball is dropped from a height of 80 metres and each bounce raises it to 60% of its previous height. (i) What height does it reach after the 5th bounce? (ii) What is the total vertical distance travelled by the time it hits the ground for the 6th time?

SOLUTION — step by step

Step 1. (i) Bounce heights: 80 × 0.6 = 48, 28.8, 17.28, 10.368, 6.2208 - after the 5th bounce the ball reaches 6.2208 m.

Step 2. (ii) Down-moves: 80 (initial drop), then 48, 28.8, 17.28, 10.368, 6.2208 before each of hits 1 to 6.

Step 3. Up-moves between hits: 48, 28.8, 17.28, 10.368, 6.2208 (the same values again).

Step 4. Total = 80 + 2 × (48 + 28.8 + 17.28 + 10.368 + 6.2208).

Step 5. Inside bracket: 48 + 28.8 = 76.8; + 17.28 = 94.08; + 10.368 = 104.448; + 6.2208 = 110.6688.

Step 6. Total = 80 + 2 × 110.6688 = 80 + 221.3376 = 301.3376 m.

» **Why this step?** Every height after the first hit is travelled TWICE (up and down); only the initial drop is travelled once - that is why the 2× bracket works.

Answer: (i) 6.2208 m (ii) 301.3376 m.

Hinglish: *5th bounce: 6.2208 m. Total distance ka funda: pehla drop ek baar (80), uske baad har upar-neeche round double. 2 × 110.6688 + 80 = 301.3376 m. 'Kitni baar travel hua' sochna is question ki asli difficulty hai.*

QUESTION 63 | Moderate | Squares as sums of odds, verified (T1/T10)

Q63. (i) Verify that 36 = 1 + 3 + 5 + 7 + 9 + 11. (ii) Write 49 and 64 as sums of consecutive odd numbers. (iii) State the general rule this illustrates.

SOLUTION — step by step

Step 1. (i) 1 + 3 = 4; + 5 = 9; + 7 = 16; + 9 = 25; + 11 = 36 ✓ - six odd numbers make 6^2.

Step 2. (ii) 49 = 1 + 3 + 5 + 7 + 9 + 11 + 13 (seven odds).

Step 3. 64 = 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 (eight odds).

Step 4. (iii) General rule: the sum of the first n odd numbers is n^2; equivalently the square sequence grows by consecutive odd numbers.

» **Why this step?** The odd numbers 1, 3, 5, ... are themselves an AP - so the square sequence is the 'running total' of an AP, which is exactly why its differences grow by 2 each time.

Answer: (ii) 49 = 1 + 3 + ... + 13; 64 = 1 + 3 + ... + 15; (iii) sum of first n odd numbers = n^2.

Hinglish: *Har square pichhle square se agla odd number jodkar banta hai: 25 + 11 = 36, 36 + 13 = 49. Bas yahi pattern hai. Rule yaad rakho - first n odd numbers ka sum = n^2.*

QUESTION 64 | Moderate | Which n gives sum 465 (T10)

Q64. The sum 1 + 2 + 3 + ... + n equals 465. Find n.

SOLUTION — step by step

Step 1. Solve $n(n + 1)/2 = 465$.

Step 2. $n(n + 1) = 930$.

Step 3. Look for consecutive factors of 930: $30 \times 31 = 930$. ✓

Step 4. So n = 30. Check: $30 \times 31/2 = 465$ ✓.

» **Why this step?** Undoing S_n means finding two consecutive numbers whose product is 2S - spotting factor pairs is faster than algebra here.

Answer: n = 30.

Hinglish: *n(n+1) = 930. Lagataar do numbers ka product 930? 30 × 31. Toh n = 30. Aise 'S_n ulta' questions mein factor pair dhoondo - quadratic solve karne se tez hai.*

QUESTION 65 | Moderate | AP property around a zero term (T8)

Q65. The 5th term of an AP is 0. Prove that $t_2 + t_8 = 0$ and $t_3 + t_7 = 0$. Does this work for any pair t_{5-k}, t_{5+k}?

SOLUTION — step by step

Step 1. Let the AP have first term a and difference d. $t_5 = a + 4d = 0$, so $a = -4d$.

Step 2. $t_2 + t_8 = (a + d) + (a + 7d) = 2a + 8d = 2(a + 4d) = 2 \times 0 = 0$. ✓

Step 3. $t_3 + t_7 = (a + 2d) + (a + 6d) = 2a + 8d = 0$. ✓

Step 4. Generally $t_{5-k} + t_{5+k} = (a + (4 - k)d) + (a + (4 + k)d) = 2a + 8d = 0$ - yes, for every k.

Step 5. Meaning: terms equidistant from a zero term cancel in pairs - e.g. -8, -5, -2, 1, ... hmm take -6, -4, -2, 0, 2, 4, 6 ($t_5 = 0$): $t_2 + t_8 = -4 + 4 = 0$ ✓.

» **Why this step?** Terms at equal distances from the middle have indices adding to the same total - their a-parts add to 2a and their d-parts cancel.

Answer: Both sums equal 0; and in general $t_{5-k} + t_{5+k} = 0$ for every k.

Hinglish: *Symmetry ka khel: zero term se jitna aage, utna peechhe - dono ka sum zero. Example list: -6, -4, -2, 0, 2, 4, 6. Ye property MCQ mein direct aa jaati hai - proof 2 lines ka hai, yaad kar lo.*

QUESTION 66 | Moderate | Shrinking GP 1, 1/3, 1/9 (T11/T12)

Q66. For the GP 1, 1/3, 1/9, ...: (i) state a and r. (ii) Find the 6th term. (iii) Which term equals 1/729?

SOLUTION — step by step

Step 1. (i) a = 1; r = (1/3)/1 = 1/3 (verify (1/9)/(1/3) = 1/3 ✓).

Step 2. $t_n = 1 \times (1/3)^{n-1} = 1/3^{n-1}$.

Step 3. (ii) $t_6 = (1/3)^5 = 1/243$.

Step 4. (iii) Set $1/3^{n-1} = 1/729$. Since $729 = 3^6$, we need n - 1 = 6, n = 7.

Step 5. 1/729 is the 7th term.

» **Why this step?** Powers of 3 to know cold: 3, 9, 27, 81, 243, 729, 2187. With them, every part of this question is a lookup.

Answer: (i) a = 1, r = 1/3 (ii) 1/243 (iii) the 7th term.

Hinglish: *Har term 3 se divide ho raha hai - r = 1/3. 6th term = 1/243. 1/729? 729 = 3^6 yaad hai toh n = 7 turant. GP mein powers ki table yaad hona superpower hai.*

QUESTION 67 | Moderate | Virahanka terms up to 144 (T6)

Q67. In the Virahānka-Fibonacci sequence 1, 2, 3, 5, 8, 13, 21, 34, ...: (i) find V_9, V_{10}, V_{11}. (ii) Which term of the sequence is 144?

SOLUTION — step by step

Step 1. (i) $V_9 = 34 + 21 = 55$.

Step 2. $V_{10} = 55 + 34 = 89$.

Step 3. $V_{11} = 89 + 55 = 144$.

Step 4. (ii) Counting the terms: $V_1 = 1$, $V_2 = 2$, $V_3 = 3$, $V_4 = 5$, $V_5 = 8$, $V_6 = 13$, $V_7 = 21$, $V_8 = 34$, $V_9 = 55$, $V_{10} = 89$, $V_{11} = 144$.

Step 5. So 144 is the 11th term of this sequence.

» **Why this step?** Because each term feeds the next two, keep a running tally of positions as you build - the Fibonacci-type list has no shortcut formula at this level.

Answer: (i) 55, 89, 144 (ii) the 11th term.

Hinglish: *55, 89, aur phir 89 + 55 = 144 - 11th term. Virahānka sequence mein terms banate jao aur position ginte jao. Ye sequence nature se lekh choka hai: flowers, shells, poetry metres - sab jagah.*

QUESTION 68 | Moderate | Find the rule then jump far (T3/T8)

Q68. A sequence begins 5, 9, 13, 17, ... (i) Assuming the obvious linear pattern, write t_n. (ii) Use it to find the 100th term.

SOLUTION — step by step

Step 1. (i) Differences: 4, 4, 4 - an AP with a = 5, d = 4.

Step 2. $t_n = 5 + (n - 1) \times 4 = 4n + 1$.

Step 3. (ii) $t_{100} = 4 \times 100 + 1 = 401$.

» **Why this step?** Constant gaps certify the AP, and then the 100th term is a single multiplication - never write 100 terms.

Answer: (i) $t_n = 4n + 1$ (ii) 401.

Hinglish: *Gap 4 constant - AP pakka. $t_n = 4n + 1$. 100th = 401. Simple hai, par exam mein log phir bhi terms likhte reh jaate hain - formula pehle, list kabhi nahi.*

QUESTION 69 | Moderate | Rule and membership together (T3/T4)

Q69. A sequence begins 7, 10, 13, 16, ... (i) Write t_n. (ii) Is 100 a term of this sequence? If yes, which one?

SOLUTION — step by step

Step 1. (i) Gaps are 3 - AP with a = 7, d = 3: $t_n = 7 + (n - 1) \times 3 = 3n + 4$.

Step 2. (ii) Solve $3n + 4 = 100$: $3n = 96$, $n = 32$.

Step 3. n = 32 is a natural number, so 100 is a term - the 32nd.

» **Why this step?** Finding the rule first turns membership into a one-line equation instead of listing 32 terms.

Answer: (i) $t_n = 3n + 4$ (ii) yes, the 32nd term.

Hinglish: *Rule banao (3n + 4), phir 100 ka position nikaalo: n = 32. Ek hi mein dono kaam - rule + membership. Listing karne wale 32 lines likh rahe honge abhi tak.*

QUESTION 70 | Moderate | Doubling money (T13)

Q70. A magical wallet doubles the money inside it every day, starting with ₹1 on day 1. (i) How much money is inside on day 15? (ii) On day 30?

SOLUTION — step by step

Step 1. Amounts: ₹1, ₹2, ₹4, ₹8, ... - a GP with a = 1, r = 2.

Step 2. Day-n amount: $t_n = 1 \times 2^{n-1} = 2^{n-1}$.

Step 3. (i) Day 15: 2^{14} = ₹16,384.

Step 4. (ii) Day 30: 2^{29} = ₹53,68,70,912 (over ₹53 crore!).

» **Why this step?** Day 1 holds the starting amount, so the exponent is n - 1, not n - the classic indexing choice in doubling problems.

Answer: (i) ₹16,384 (ii) ₹53,68,70,912.

Hinglish: *15 din mein ₹16,384 aur 30 din mein 53 crore se zyada! Doubling ki taakat yahi hai - exponent har din +1. Day 1 pe starting amount hai isliye 2^{n-1}, off-by-one dhyan se.*

QUESTION 71 | Moderate | Auditorium seating (T8/T10)

Q71. An auditorium has 18 seats in the first row, 20 in the second, 22 in the third, and so on for 15 rows. (i) How many seats are in the last row? (ii) How many seats does the auditorium have in total?

SOLUTION — step by step

Step 1. (i) Row counts: 18, 20, 22, ... - an AP with a = 18, d = 2.

Step 2. 15th row: t_{15} = 18 + 14 × 2 = 46 seats.

Step 3. (ii) Total = (first + last)/2 × number of rows = (18 + 46)/2 × 15.

Step 4. = 64/2 × 15 = 32 × 15 = 480.

» **Why this step?** The average-row size times the number of rows counts the seats - Āryabhaṭa's average trick applied to seating.

Answer: (i) 46 seats (ii) 480 seats.

Hinglish: *Last row: 18 + 14 steps × 2 = 46. Total: average 32 × 15 rows = 480. Row-wise jodne ke chakkar mein time waste mat karo - pehla + aakhira, aadha, guna rows.*

QUESTION 72 | Moderate | Snail on a pole - the reach-day twist (T14)

Q72. A snail climbs a 20-metre pole. Each day it climbs 5 metres; each night it slips down 2 metres. On which day does it reach the top of the pole?

SOLUTION — step by step

Step 1. Net gain per full day-night cycle: 5 - 2 = 3 m (as long as it has not reached the top).

Step 2. End of day 5 (after 5 nights): height = 5 × 3 = 15 m.

Step 3. On day 6 the snail climbs 5 m from 15 m: 15 + 5 = 20 m - it reaches the top during day 6, BEFORE any night slip can occur.

Step 4. Trap check: '20/3 = 6.67, so day 7' is wrong - the final climb needs no subtraction for slipping because the snail stops at the top.

» **Why this step?** Cycle problems always have a 'last-day special case': the final upward climb finishes the job, so divide (target - final climb) by the net rate, then add the last day.

Answer: Day 6.

Hinglish: *Roz net +3, par last day pe sarak nahi lagti. 15 m tak 5 din, phir 6th day ke climb se seedha top. '20/3 = 6.67 toh 7th day' - ye galti hai, kyunki aakhri din slip ka mauka hi nahi milta.*

QUESTION 73 | Moderate | Paper folding explosion (T13)

Q73. A sheet of paper is 0.1 mm thick. Each fold doubles its thickness. (i) What is the thickness after 20 folds? (ii) Compare it with the Qutub Minar (about 73 m tall).

SOLUTION — step by step

Step 1. Thickness after n folds: 0.1×2^n mm (fold 1 gives 0.2 mm, fold 2 gives 0.4 mm, ...).

Step 2. (i) $2^{20} = 1{,}048{,}576$.

Step 3. Thickness = $0.1 \times 1{,}048{,}576 = 1{,}04{,}857.6$ mm = 104.8576 m.

Step 4. (ii) 104.86 m is about 1.44 times the Qutub Minar's height - a sheet thinner than your hair becomes taller than a monument in 20 folds!

» **Why this step?** Twenty doublings multiply by about a million - exponential growth defeats all human intuition about 'just one more fold'.

Answer: (i) 104.8576 m (ii) about 1.4 times taller than the Qutub Minar.

Hinglish: *20 folds = 2^{20} = 10 lakh se zyada! 0.1 mm ka paper 104.86 metre - Qutub Minar se ooncha. GP ka kamaal: chhota multiplier, bada exponent - tabahi.*

QUESTION 74 | Moderate | Ten percent yearly growth (T13)

Q74. Meera's starting salary is ₹2,40,000 per year, and it rises by 10% every year. (i) Show that her yearly salaries form a GP and state a and r. (ii) Find her salary in the 3rd year.

SOLUTION — step by step

Step 1. (i) A 10% rise multiplies each salary by 1.10: 2,40,000; 2,64,000; 2,90,400; ...

Step 2. Check ratio: 2,64,000/2,40,000 = 1.1 and 2,90,400/2,64,000 = 1.1 - a GP with a = 2,40,000, r = 1.1.

Step 3. $t_n = 2{,}40{,}000 \times (1.1)^{n-1}$.

Step 4. (ii) 3rd year: $2{,}40{,}000 \times (1.1)^2 = 2{,}40{,}000 \times 1.21 =$ ₹2,90,400.

» **Why this step?** 'r% increase' always means r_GP = 1 + r/100; a 10% hike is a multiplier of 1.1, not an addition of the same amount every year.

Answer: (i) GP with a = 2,40,000, r = 1.1 (ii) ₹2,90,400.

Hinglish: *10% badhna = har baar 1.1 se multiply - GP. Increment wale Harish wale question se farak dekho: wahan SAME ₹20,000 judd rahe the (AP), yahan SAME 10% multiply ho raha hai (GP). Salary +10% ka matlab 1.1 ka factor.*

QUESTION 75 | Moderate | Chessboard rice (T12/T13)

Q75. On a chessboard, 1 grain of rice is placed on the first square, 2 on the second, 4 on the third, doubling up to the 64th square. How many grains are on the 20th square?

SOLUTION — step by step

Step 1. Counts: 1, 2, 4, 8, ... - a GP with a = 1, r = 2.

Step 2. Square-n count: $t_n = 2^{n-1}$ (square 1 holds $1 = 2^0$).

Step 3. 20th square: $2^{19} = 5{,}24{,}288$ grains.

» **Why this step?** Square number = exponent + 1: the 20th square is 2^{19}, not 2^{20}.

Answer: 5,24,288 grains.

Hinglish: *Chessboard ka classic doubling. 20th square = 2^{19} = 5,24,288 grains - aur ye toh aadha board bhi nahi hua! 64th square par grains ka pahad ban jaata hai.*

QUESTION 76 | Moderate | Inserting terms in an AP (T8/T9)

Q76. Insert three numbers between 5 and 21 so that the five numbers form an arithmetic progression. Write the full AP.

SOLUTION — step by step

Step 1. Five AP terms: 5, t_2, t_3, t_4, 21.

Step 2. The 5th term is 21: a + 4d = 21 with a = 5, so 5 + 4d = 21.

Step 3. 4d = 16, so d = 4.

Step 4. Terms: 5, 9, 13, 17, 21.

Step 5. Check: gaps are 4 everywhere and the last term is 21 ✓.

» **Why this step?** Between two endpoints, k inserted terms create k + 1 gaps of equal size d = (last - first)/(k + 1) - here (21 - 5)/4 = 4.

Answer: 9, 13, 17 - the AP is 5, 9, 13, 17, 21.

Hinglish: *5 aur 21 ke beech 3 numbers = 4 equal gaps. d = 16/4 = 4. Toh 9, 13, 17 ghusao. Formula yaad rakho: (last - first) ÷ (kitne gaps) = d.*

Level 3 · Exam Power Zone: long and twisted:

QUESTION 77 | Difficult | 31st term from t11 and t16 (T9)

Q77. Find the 31st term of an AP whose 11th term is 38 and whose 16th term is 73.

SOLUTION — step by step

Step 1. a + 10d = 38 and a + 15d = 73.

Step 2. Subtract: 5d = 35, so d = 7.

Step 3. Back-substitute: a = 38 - 10 × 7 = 38 - 70 = -32.

Step 4. t_{31} = a + 30d = -32 + 30 × 7 = -32 + 210 = 178.

» **Why this step?** Two term-facts minus each other give (difference of indices) × d - here 5 steps apart means 5d = 35.

Answer: 178.

Hinglish: *11th aur 16th ke beech 5 steps = 5d = 35, d = 7. a negative nikla (-32) - koi problem nahi, AP negative se shuru hokar chadhti hai. 31st = 178.*

QUESTION 78 | Difficult | AP from t3 and a gap condition (T9)

Q78. Determine the AP whose 3rd term is 16 and whose 7th term exceeds the 5th term by 12.

SOLUTION — step by step

Step 1. $t_7 - t_5$ = (a + 6d) - (a + 4d) = 2d = 12, so d = 6.

Step 2. t_3 = a + 2d = 16 gives a + 12 = 16, so a = 4.

Step 3. The AP is 4, 10, 16, 22, 28, ...

Step 4. Check: t_7 = 4 + 36 = 40, t_5 = 4 + 24 = 28, and 40 - 28 = 12 ✓.

» **Why this step?** A condition comparing two terms (exceeds by...) is a DIFFERENCE of two nth-terms - it pins down d alone, with no need for a.

Answer: The AP is 4, 10, 16, 22, ... (a = 4, d = 6).

Hinglish: *'7th, 5th se 12 zyada hai' = 2d = 12 = seedha d mil gaya. a t_3 se nikala. Gap conditions hamesha d deti hain - pehle woh karo, a baad mein.*

QUESTION 79 | Difficult | Three-digit multiples of 7 (T8)

Q79. How many three-digit numbers are divisible by 7?

SOLUTION — step by step

Step 1. Smallest three-digit multiple of 7: 7 × 14 = 98 is two-digit; 7 × 15 = 105 - so start at 105.

Step 2. Largest: 7 × 142 = 994 (7 × 143 = 1001 is four-digit).

Step 3. The list 105, 112, 119, ..., 994 is an AP with d = 7.

Step 4. Solve 105 + (n - 1) × 7 = 994: (n - 1) × 7 = 889, n - 1 = 127, n = 128.

Step 5. There are 128 three-digit multiples of 7.

» **Why this step?** Locate the endpoints by dividing the boundary numbers by 7, then count with the nth-term formula - no listing needed.

Answer: 128.

Hinglish: *105 se 994 tak, step 7. Count = (994 - 105)/7 + 1 = 128. Endpoints nikalna hi asli kaam hai - 98 aur 1001 ko dhakke maar ke bahar karo.*

QUESTION 80 | Difficult | Multiples of 4 strictly between 10 and 250 (T8)

Q80. How many multiples of 4 lie between 10 and 250?

SOLUTION — step by step

Step 1. Smallest multiple of 4 above 10: 12 (4 × 3).

Step 2. Largest multiple of 4 below 250: 248 (4 × 62).

Step 3. The AP 12, 16, 20, ..., 248 has d = 4.

Step 4. Count: 12 + (n - 1) × 4 = 248 gives (n - 1) × 4 = 236, n - 1 = 59, n = 60.

Step 5. So 60 multiples of 4 lie strictly between 10 and 250.

» **Why this step?** 'Between' means strict: 10 and 250 themselves are not in the list (and are not multiples of 4 anyway). Snap to the nearest inside multiples first.

Answer: 60.

Hinglish: *12 se 248 tak, step 4: (248 - 12)/4 + 1 = 60. Boundary numbers ko pehle andar waale multiples se replace karo, phir counting formula.*

QUESTION 81 | Difficult | GP from two conditions (T12)

Q81. Find a GP in which the sum of the first two terms is -4 and the 5th term is 4 times the 3rd term.

SOLUTION — step by step

Step 1. Let the GP be a, ar, ar^2, ... with r not equal to 0.

Step 2. Condition 2: $ar^4 = 4ar^2$. Divide by ar^2: $r^2 = 4$, so r = 2 or r = -2.

Step 3. Condition 1: a + ar = a(1 + r) = -4.

Step 4. If r = 2: a(3) = -4, a = -4/3. GP: -4/3, -8/3, -16/3, -32/3, -64/3, ... Check: $t_5 = -64/3 = 4 \times (-16/3)$ ✓.

Step 5. If r = -2: a(1 - 2) = -4, so -a = -4, a = 4. GP: 4, -8, 16, -32, 64, ... Check: $t_5 = 64 = 4 \times 16 = 4 \times t_3$ ✓.

Step 6. Both GPs satisfy the conditions - the answer is not unique.

» **Why this step?** The 5th = 4 × 3rd condition gives $r^2 = 4$, so TWO ratios qualify - a complete answer must report both branches.

Answer: r = 2 with a = -4/3: (-4/3, -8/3, -16/3, ...); or r = -2 with a = 4: (4, -8, 16, -32, 64, ...).

Hinglish: *$r^2 = 4$ se do r nikle: 2 aur -2. Dono ke liye alag-alag a. Do GP valid hain - sirf ek likhna incomplete answer hai. Squares hamesha do roots dete hain, dono check karo.*

QUESTION 82 | Difficult | 100 as sums of consecutive naturals (T10)

Q82. Find all possible ways of expressing 100 as a sum of two or more consecutive natural numbers.

SOLUTION — step by step

Step 1. Suppose 100 = (k + 1) + (k + 2) + ... + (k + m), m consecutive numbers starting after k.

Step 2. Then $100 = S_{k+m} - S_k = (k + m)(k + m + 1)/2 - k(k + 1)/2$.

Step 3. Test small counts m with the average idea: the average 100/m must sit between consecutive positions.

Step 4. m = 5: average 20 - terms 18 + 19 + 20 + 21 + 22 = 100 ✓.

Step 5. m = 8: average 12.5 - terms 9 + 10 + ... + 16 = 100 ✓ (check: 8 × 12.5 = 100).

Step 6. m = 2: average 50 - would need 49.5 + 50.5, not natural. m = 4: average 25 - needs half-integers 23.5...26.5. m = 3: 33.33 - no. m = 6: 16.67 - no. m = 7: 14.29 - no. Larger m push the start at or below 0. So only two ways exist.

» **Why this step?** m consecutive naturals sum to m × (their average); a valid split needs the average to be a half-integer that lands on whole terms.

Answer: 100 = 18 + 19 + 20 + 21 + 22 = 9 + 10 + 11 + 12 + 13 + 14 + 15 + 16 - exactly two ways.

Hinglish: *18 se 22 tak (5 terms, avg 20) aur 9 se 16 tak (8 terms, avg 12.5) - bas yehi do tareeke. Baaki m ke liye average integer ya .5 wala nahi aata. 'All possible ways' poochha hai toh chhote m systematically try karo.*

QUESTION 83 | Difficult | AP from two sums of terms (T9)

Q83. The sum of the 4th and 8th terms of an AP is 24, and the sum of the 6th and 10th terms is 44. Find the first three terms of the AP.

SOLUTION — step by step

Step 1. $t_4 + t_8 = (a + 3d) + (a + 7d) = 2a + 10d = 24$.

Step 2. $t_6 + t_{10} = (a + 5d) + (a + 9d) = 2a + 14d = 44$.

Step 3. Subtract: $4d = 20$, so $d = 5$.

Step 4. Then $2a = 24 - 50 = -26$, so $a = -13$.

Step 5. First three terms: -13, -8, -3. Check: $t_4 = 2$, $t_8 = 22$, sum 24 ✓; $t_6 = 12$, $t_{10} = 32$, sum 44 ✓.

» **Why this step?** Sums of symmetric-ish term pairs collapse to 2a + (fixed multiple of d) - subtract the two equations and d falls out.

Answer: -13, -8, -3.

Hinglish: *Dono sums ko a, d mein kholo, minus karo: 4d = 20. a = -13 nikla - negative start, koi baat nahi. Verify zaroor karo dono conditions - full marks ka proof.*

QUESTION 84 | Difficult | Smallest n with S-sub-n beyond 1000 (T10)

Q84. Find the smallest value of n such that the sum of the first n natural numbers is greater than 1,000.

SOLUTION — step by step

Step 1. $S_n = n(n + 1)/2 > 1000$ means $n(n + 1) > 2000$.

Step 2. Estimate: $44 \times 45 = 1980$ (just short); $45 \times 46 = 2070$ (over).

Step 3. So $n(n + 1) > 2000$ first happens at $n = 45$.

Step 4. Verify: $S_{44} = 1980/2 = 990$ (not enough); $S_{45} = 2070/2 = 1035 > 1000$ ✓.

» **Why this step?** Scan factor pairs near the square root of 2×1000: the crossing point is where $n(n + 1)$ sneaks past 2000.

Answer: n = 45 ($S_{45} = 1035$ is the first sum past 1,000).

Hinglish: *n(n+1) ko 2000 se upar dhakelna hai. 44 × 45 = 1980 - fusss, kam. 45 × 46 = 2070 - ho gaya. Toh n = 45, sum 1035. Aise 'smallest n' questions mein do neighbouring values check karna hi proof hai.*

QUESTION 85 | Difficult | Which term of 2, 8, 32 is 131072 (T12)

Q85. Which term of the GP 2, 8, 32, ... is 131072? Write the explicit and recursive formulas for the nth term.

SOLUTION — step by step

Step 1. a = 2; r = 8/2 = 4 (verify 32/8 = 4 ✓).

Step 2. Explicit: $t_n = 2 \times 4^{n-1} = 2 \times 2^{2n-2} = 2^{2n-1}$.

Step 3. Set $2^{2n-1} = 131072$. Powers: $2^{17} = 131072$.

Step 4. 2n - 1 = 17 gives n = 9.

Step 5. So 131072 is the 9th term. Recursive formula: $t_1 = 2$, $t_n = 4t_{n-1}$.

Step 6. Quick check: 2, 8, 32, 128, 512, 2048, 8192, 32768, 131072 ✓.

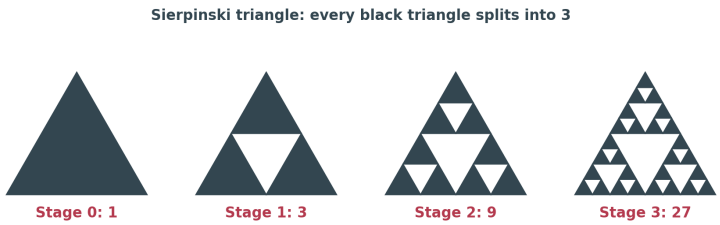
» **Why this step?** Reduce everything to one base: $4 = 2^2$ turns the GP into pure powers of 2, and $131072 = 2^{17}$ seals it.

Answer: The 9th term; explicit $t_n = 2^{2n-1}$; recursive $t_1 = 2$, $t_n = 4t_{n-1}$.

Hinglish: *Sab 2 ki powers mein likho: 2, 2^3, 2^5... odd powers chal rahe hain. 131072 = 2^{17}, toh 2n - 1 = 17, n = 9. Powers of 2 yaad karo 2^{10} = 1024 tak, aage double karte jao.*

QUESTION 86 | Difficult | Sierpinski triangle counts and areas (T13)

Q86. The figure shows Stages 0 to 3 of the Sierpiński triangle, built from an equilateral triangle by repeatedly joining midpoints and removing the central triangle. (i) How many black triangles are there at Stages 0 to 3? (ii) Predict Stages 4 and 5 and find the rule for Stage n. (iii) If the Stage-0 black area is 1 square unit, what is the black area at Stage n, and what happens to it as n grows?



SOLUTION — step by step

Step 1. (i) Stage 0: 1 black triangle; Stage 1: 3; Stage 2: 9; Stage 3: 27 - every black triangle is replaced by 3 smaller ones.

Step 2. (ii) Stage 4: 81; Stage 5: 243. Rule: at Stage n the count is 3^n - a GP with a = 1, r = 3 indexed from Stage 0.

Step 3. (iii) Each stage keeps 3 of every 4 equal pieces, so the area multiplies by 3/4: area at Stage n = $(3/4)^n$ (1, 3/4, 9/16, 27/64, ...).

Step 4. As n grows, $(3/4)^n$ heads to 0 - the number of triangles explodes while the total black area melts away.

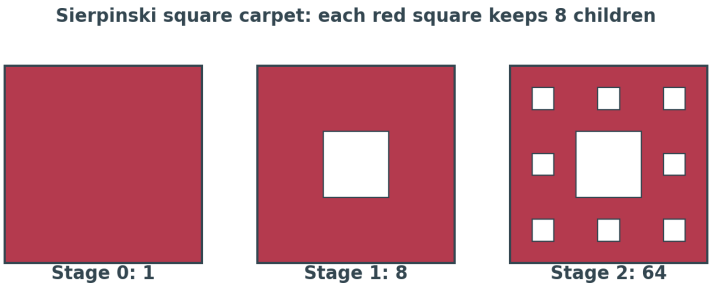
» **Why this step?** Count and area ride two different GPs: x3 for the count, x3/4 for the area. Stage indexing starts at 0 - that is why the count is 3^n and not 3^{n-1}.

Answer: (i) 1, 3, 9, 27 (ii) 81 and 243; count at Stage n = 3^n (iii) area = $(3/4)^n$, which tends to 0.

Hinglish: *Count har stage par 3 se multiply: 1, 3, 9, 27, 81, 243 - Stage n pe 3^n. Area har baar 3/4 ho jaata hai: $(3/4)^n$. Ek cheez badhti hai, doosri ghatti hai - fractal ka jaadu. Stage 0 se shuru - exponent mein +1 ka farak yaad rakho.*

QUESTION 87 | Difficult | Sierpinski carpet counts and areas (T13)

Q87. In the Sierpiński square carpet (the figure shows Stages 0 to 2), each square sheet is divided into a 3 × 3 grid and the centre square is removed, then the process repeats on every remaining square. (i) How many red squares remain at Stages 0, 1, 2? (ii) Find the rule for Stage n and predict Stage 4. (iii) If the Stage-0 area is 1 square unit, find the red area at Stage n and at Stage 2 exactly.



SOLUTION — step by step

Step 1. (i) Stage 0: 1 red square. Stage 1: each sheet keeps 8 of 9 pieces, so 8. Stage 2: 8 × 8 = 64.

Step 2. (ii) Count at Stage n = 8^n (a GP with a = 1, r = 8, indexed from Stage 0). Stage 4: 8^4 = 4096.

Step 3. (iii) Area keeps 8/9 of itself each stage: area at Stage n = $(8/9)^n$.

Step 4. Stage 2 area: $(8/9)^2$ = 64/81 square units.

» **Why this step?** Same fractal logic as the triangle, new numbers: 9 pieces keep 8, so count x8 and area x8/9 per stage.

Answer: (i) 1, 8, 64 (ii) 8^n; Stage 4 = 4096 (iii) area = $(8/9)^n$; Stage 2 = 64/81.

Hinglish: *3×3 grid, beech wala kata - 8 bachche har square ke. Count 8^n: 1, 8, 64, 512, 4096. Area $(8/9)^n$: 1, 8/9, 64/81... Triangle wale fractal se sirf numbers badle hain, kahani wahi.*

QUESTION 88 | Difficult | Sierpinski counting drill (T13)

Q88. For the Sierpiński triangle: (i) how many black triangles are there at Stage 6? (ii) At which stage are there exactly 6561 black triangles? (iii) What is the black area at Stage 4 if Stage 0 has area 1?

SOLUTION — step by step

Step 1. (i) Count at Stage n = 3^n, so Stage 6: $3^6 = 729$.

Step 2. (ii) Solve $3^n = 6561$. Since $3^8 = 6561$ ($3^4 = 81$, $3^8 = 81^2 = 6561$), the stage is n = 8.

Step 3. (iii) Area at Stage n = $(3/4)^n$, so Stage 4: $(3/4)^4 = 81/256$ square units.

» **Why this step?** Know powers of 3 up to 3^8: 3, 9, 27, 81, 243, 729, 2187, 6561. Fractal stage questions become instant lookups.

Answer: (i) 729 (ii) Stage 8 (iii) 81/256 square units.

Hinglish: *729 triangles at Stage 6. 6561? 81 ka square = 3^8, toh Stage 8. Area: $(3/4)^4 = 81/256$. Powers of 3 ka chart bana lo - 3 se 6561 tak - fractal questions phir 10 second ke hain.*

QUESTION 89 | Difficult | GP from t2 and t5 (T12)

Q89. In a GP the 2nd term is 9 and the 5th term is 243. Find the GP and its 8th term.

SOLUTION — step by step

Step 1. $ar = 9$ and $ar^4 = 243$.

Step 2. Divide: $ar^4/ar = r^3 = 243/9 = 27$, so r = 3.

Step 3. $a = 9/r = 3$.

Step 4. GP: 3, 9, 27, 81, 243, ... (check: $t_5 = 3 \times 3^4 = 243$ ✓).

Step 5. $t_8 = 3 \times 3^7 = 3^8 = 6561$.

» **Why this step?** Dividing two term-equations kills a: r appears to the power (index difference) = 3 here. Powers of 3 do the rest.

Answer: GP: 3, 9, 27, 81, ...; $t_8 = 6561$.

Hinglish: *Do equations divide karo - a gayab, $r^3 = 27$, r = 3. wapas daal ke a = 3. 8th term = $3^8 = 6561$. GP ke two-condition questions ka standard recipe yahi hai.*

QUESTION 90 | Difficult | GP with negative ratio from t2, t5 (T12)

Q90. In a GP the 2nd term is 18 and the 5th term is -486. Find the GP and its 7th term.

SOLUTION — step by step

Step 1. ar = 18 and ar^4 = -486.

Step 2. Divide: r^3 = -486/18 = -27, so r = -3 (a negative ratio explains the sign flip).

Step 3. a = 18/r = -6.

Step 4. GP: -6, 18, -54, 162, -486, ... (check: $t_5 = -6 \times (-3)^4 = -6 \times 81 = -486$ ✓).

Step 5. $t_7 = -6 \times (-3)^6 = -6 \times 729 = -4374$.

» **Why this step?** A negative r^3 forces r negative: odd powers keep the sign of r, and here the 5th term's minus sign is the clue.

Answer: GP: -6, 18, -54, 162, -486, ...; t_7 = -4374.

Hinglish: *-486/18 = -27 = r^3 - negative cube, toh r = -3 pakka. Terms baari-baari sign badalti hain: -6, 18, -54... t_7 = -4374. Sign se r ke baare mein pehli clue milti hai - use karo.*

QUESTION 91 | Difficult | First negative term of a decreasing AP (T8)

Q91. Consider the AP 100, 92, 84, 76, ... Which term of this AP is the FIRST negative term?

SOLUTION — step by step

Step 1. a = 100, d = -8. $t_n = 100 - 8(n - 1) = 108 - 8n$.

Step 2. We need the smallest n with $t_n < 0$: $108 - 8n < 0$.

Step 3. $8n > 108$, so $n > 13.5$.

Step 4. Smallest whole n = 14.

Step 5. $t_{13} = 108 - 104 = 4$ (still positive); $t_{14} = 108 - 112 = -4$ - the first negative term is the 14th, equal to -4.

» **Why this step?** Solve the inequality, then round UP to the next whole position - and display both neighbours to prove the crossing.

Answer: The 14th term, which equals -4.

Hinglish: *$108 - 8n < 0$ se $n > 13.5$, toh 14th term. 13th = +4, 14th = -4 - crossing yahin hui. 'First negative' = inequality solve karo, ceiling lo, dono terms dikhao.*

QUESTION 92 | Difficult | Falling stone distances (T8/T10)

Q92. A stone falling from rest covers 4.9 m in the 1st second, 9.8 m in the 2nd second, 14.7 m in the 3rd second, and so on - each second 4.9 m more than the previous second. (i) How far does it fall during the 10th second? (ii) How far in total during the first 10 seconds?

SOLUTION — step by step

Step 1. (i) Per-second distances: 4.9, 9.8, 14.7, ... - an AP with a = 4.9, d = 4.9.

Step 2. 10th-second distance: $t_{10} = 4.9 + 9 \times 4.9 = 4.9 \times 10 = 49$ m.

Step 3. (ii) Total = $4.9 + 9.8 + ... + 49 = 4.9 \times (1 + 2 + ... + 10) = 4.9 \times 55$.

Step 4. = 269.5 m.

Step 5. Cross-check with average × count: $(4.9 + 49)/2 \times 10 = 26.95 \times 10 = 269.5$ ✓.

» **Why this step?** Factor out the common 4.9: the remaining list is 1 + 2 + ... + 10 - the natural-number sum wearing a disguise.

Answer: (i) 49 m (ii) 269.5 m.

Hinglish: *Har second 4.9 extra - AP. 10th second = 4.9 × 10 = 49 m. Total = 4.9 × (1+...+10) = 4.9 × 55 = 269.5 m. Common factor bahar nikaalo, andar S_{10} dikh jaata hai.*

QUESTION 93 | Difficult | Sum of three-digit multiples of 9 (T8/T10)

Q93. Find the sum of all three-digit numbers divisible by 9.

SOLUTION — step by step

Step 1. Endpoints: smallest 108 (9 × 12), largest 999 (9 × 111).

Step 2. Count: (999 - 108)/9 + 1 = 891/9 + 1 = 99 + 1 = 100 numbers.

Step 3. Sum = (first + last)/2 × count = (108 + 999)/2 × 100.

Step 4. = 1107/2 × 100 = 553.5 × 100 = 55350.

» **Why this step?** First-last-average × count handles any 'sum all multiples of k' question - endpoints and count are the only real work.

Answer: 55350.

Hinglish: *108 se 999, 100 numbers. Average (108 + 999)/2 = 553.5, × 100 = 55350. Count mein +1 aur average mein .5 - dono se ghabrana nahi, multiplication sab theek kar deta hai.*

QUESTION 94 | Difficult | Recursion that builds squares (T5/T14)

Q94. A sequence is defined by $u_1 = 1$ and $u_n = u_{n-1} + (2n - 1)$ for n at least 2. (i) Find the first five terms. (ii) Guess the explicit rule and prove it.

SOLUTION — step by step

Step 1. (i) $u_1 = 1$; $u_2 = 1 + 3 = 4$; $u_3 = 4 + 5 = 9$; $u_4 = 9 + 7 = 16$; $u_5 = 16 + 9 = 25$.

Step 2. (ii) The terms are perfect squares: $u_n = n^2$.

Step 3. Proof: $u_n = 1 + 3 + 5 + ... + (2n - 1)$ - the sum of the first n odd numbers.

Step 4. These n odd numbers are equally spaced, so their sum = (first + last)/2 × count = (1 + (2n - 1))/2 × n = (2n/2) × n = n × n = n^2. Hence $u_n = n^2$ for every n. ✓

» **Why this step?** The recursion adds consecutive odd numbers - and the average of the first n odd numbers is exactly n (first + last)/2 = (1 + 2n - 1)/2 = n, so the sum is n × n = n^2.

Answer: (i) 1, 4, 9, 16, 25 (ii) $u_n = n^2$; proof via average of the first n odd numbers = n, so sum = n × n.

Hinglish: *Terms: 1, 4, 9, 16, 25 - squares! Kyunki hum odd numbers jod rahe hain aur first n odds ka sum n^2 hota hai (average n × count n). Ye recursion square numbers ka recursive birthday hai.*

QUESTION 95 | Difficult | Recursion that builds triangular numbers (T5/T10)

Q95. A sequence is defined by $v_1 = 1$ and $v_n = v_{n-1} + n$ for n at least 2. (i) Find the first five terms. (ii) Guess and verify the explicit rule.

SOLUTION — step by step

Step 1. (i) $v_1 = 1$; $v_2 = 1 + 2 = 3$; $v_3 = 3 + 3 = 6$; $v_4 = 6 + 4 = 10$; $v_5 = 10 + 5 = 15$.

Step 2. (ii) These are the triangular numbers: $v_n = n(n + 1)/2$.

Step 3. Verify n = 5: 5 × 6/2 = 15 ✓; n = 4: 4 × 5/2 = 10 ✓.

Step 4. Why it works: $v_n = 1 + 2 + ... + n$ because each step adds the next natural number - and that sum is n(n + 1)/2.

» **Why this step?** The recursion 'add n' literally writes 1 + 2 + ... + n - the recursive and the summation views are the same sequence.

Answer: (i) 1, 3, 6, 10, 15 (ii) $v_n = n(n + 1)/2$, verified.

Hinglish: *1, 3, 6, 10, 15 - triangular numbers! Har baar agla natural number jud raha hai, toh sum formula hi explicit rule hai. Recursion dekho, pattern pehchano, formula likho - yehi game hai.*

QUESTION 96 | Difficult | Three numbers in AP: sum and product (T9)

Q96. Three numbers are in AP. Their sum is 27 and their product is 504. Find the numbers.

SOLUTION — step by step

Step 1. Let the numbers be 9 - x, 9, 9 + x (middle term = average = 27/3 = 9).

Step 2. Product: $(9 - x) \times 9 \times (9 + x) = 9(81 - x^2) = 504$.

Step 3. $81 - x^2 = 56$, so $x^2 = 25$, $x = 5$ (or -5, which only reverses the order).

Step 4. Numbers: 4, 9, 14. Check: sum 27 ✓; product $4 \times 9 \times 14 = 504$ ✓.

» **Why this step?** Writing an AP as (m - x, m, m + x) converts 'three in AP' into ONE unknown - the single smartest substitution for such problems.

Answer: 4, 9, 14 (or the same three in reverse order).

Hinglish: *Teen AP numbers ka middle = average = 9. Baaki do ko 9 - x, 9 + x likho - sum apne aap sambhal jaata hai. Product equation se $x^2 = 25$. $x = \pm 5$ dono same set dete hain, bas order ulta.*

QUESTION 97 | Difficult | Three numbers in AP: second pair (T9)

Q97. Three numbers are in AP. Their sum is 24 and their product is 440. Find the numbers.

SOLUTION — step by step

Step 1. Middle term = sum/3 = 8. Write the numbers as 8 - x, 8, 8 + x.

Step 2. Product: $8(8 - x)(8 + x) = 8(64 - x^2) = 440$.

Step 3. $64 - x^2 = 55$, so $x^2 = 9$, $x = 3$.

Step 4. Numbers: 5, 8, 11. Check: sum 24 ✓; product $5 \times 8 \times 11 = 440$ ✓.

» **Why this step?** Same (m - x, m, m + x) substitution: the sum condition vanishes automatically and only the product equation remains.

Answer: 5, 8, 11.

Hinglish: *Middle = 24/3 = 8. Product equation se $x = 3$. Numbers 5, 8, 11. Ye substitution teen-number AP questions ka master key hai - isse linear equations ka jhanjhat khatam.*

QUESTION 98 | Difficult | Angles of a quadrilateral in AP (T9)

Q98. The four angles of a quadrilateral are in AP with common difference 12 degrees. Find all four angles.

SOLUTION — step by step

Step 1. Let the angles be a, a + d, a + 2d, a + 3d with d = 12 degrees.

Step 2. Angle sum of a quadrilateral = 360 degrees: 4a + 6d = 360.

Step 3. 4a + 72 = 360, so 4a = 288, a = 72 degrees.

Step 4. Angles: 72, 84, 96, 108 degrees. Check: they increase by 12 and total 360 ✓.

» **Why this step?** The sum a + (a + d) + (a + 2d) + (a + 3d) = 4a + 6d - memorise this pattern; it appears for any 4-term AP with a fixed total.

Answer: 72, 84, 96 and 108 degrees.

Hinglish: *Chaar angles ka total 360 - AP form mein 4a + 6d = 360. d = 12 daalo, a = 72. Check karo: 72 + 84 + 96 + 108 = 360. Geometry + sequences ka combo question.*

QUESTION 99 | Difficult | Message chain GP (T13)

Q99. A student forwards an interesting fact to 3 friends. Each of them forwards it to 3 new friends in the next round, and the pattern continues, with no one receiving it twice. (i) How many NEW people receive it in round 8? (ii) How many people in total have received it by the end of round 5 (counting the student as round 1)?

SOLUTION — step by step

Step 1. (i) New receivers per round: 3, 9, 27, 81, ... - a GP with a = 3, r = 3 (round 1 = the original student).

Step 2. Round 8 new receivers = $3^7 = 2187$.

Step 3. (ii) Total by end of round 5 = $3^0 + 3^1 + 3^2 + 3^3 + 3^4 = 1 + 3 + 9 + 27 + 81 = 121$.

» **Why this step?** Round indexing: the student is round 1 with 1 person, so round n has 3^{n-1} new people - watch what 'round 8' means before computing.

Answer: (i) 2187 (ii) 121 people.

Hinglish: *Round 8 mein naye log = $3^7 = 2187$. Total 5 rounds tak = 121. Virahanka-Fibonacci ke saath compare karo: wahan jud rahe the, yahan multiply - GP ki speed ka koi muqabla nahi.*

QUESTION 100 | Difficult | Sweet distribution with AP sum (T10)

Q100. A housing society has 20 houses in a row. The first house gets 5 laddoos from a festival box, and each next house gets 3 laddoos more than the previous one. How many laddoos does the festival box need in total?

SOLUTION — step by step

Step 1. Laddoos per house: 5, 8, 11, ..., - an AP with a = 5, d = 3, n = 20.

Step 2. Last house: $t_{20} = 5 + 19 \times 3 = 62$.

Step 3. Total = (first + last)/2 × count = (5 + 62)/2 × 20.

Step 4. = (67/2) × 20 = 67 × 10 = 670.

» **Why this step?** Average × count again: 20 houses receive an average of 33.5 laddoos each.

Answer: 670 laddoos.

Hinglish: *Last house ko 62 laddoo. Average 33.5 × 20 = 670. AP sum ka formula (a + l)/2 × n - ek line ka kaam, bas last term nikalna na bhoolo.*

QUESTION 101 | Difficult | Ball with halving bounces (T13)

Q101. A ball is dropped from 16 m and each bounce returns it to half its previous height. After which bounce does its peak height first fall below 0.5 m?

SOLUTION — step by step

Step 1. Peak heights: 8, 4, 2, 1, 0.5, 0.25, ... after bounces 1, 2, 3, 4, 5, 6, ...

Step 2. Height after bounce n = $16 \times (1/2)^n$.

Step 3. Need $16 \times (1/2)^n < 0.5$, i.e. $(1/2)^n < 1/32$, i.e. $2^n > 32$.

Step 4. $2^5 = 32$ is not greater, so n must be at least 6.

Step 5. Check: bounce 5 gives exactly 0.5 m (not below); bounce 6 gives 0.25 m < 0.5 ✓. Answer: after the 6th bounce.

» **Why this step?** 'First falls below' needs strict inequality - the boundary value 0.5 itself does not count, so the answer is 6, not 5.

Answer: After the 6th bounce (peak 0.25 m).

Hinglish: *16, 8, 4, 2, 1, 0.5, 0.25... 5th bounce par exactly 0.5 - 'below' nahi hua. 6th bounce = 0.25 - ab below. Strict inequality ka farak yahi ek step banata hai.*

QUESTION 102 | Difficult | Which term of a growing fraction GP (T12)

Q102. Consider the GP 1/8, 1/4, 1/2, 1, 2, ... Which term of this GP equals 256?

SOLUTION — step by step

Step 1. a = 1/8; r = (1/4)/(1/8) = 2.

Step 2. $t_n = (1/8) \times 2^{n-1} = 2^{n-1}/2^3 = 2^{n-4}$.

Step 3. Set $2^{n-4} = 256 = 2^8$.

Step 4. n - 4 = 8, so n = 12.

Step 5. 256 is the 12th term.

» **Why this step?** Put every quantity on the base 2: $a = 2^{-3}$, then $t_n = 2^{n-4}$ and matching exponents finishes the job.

Answer: The 12th term.

Hinglish: *$1/8 = 2^{-3}$ - negative exponents se kaam aasan ho jaata hai. $t_n = 2^{n-4}$. $256 = 2^8$ toh n = 12. Base ek karo, exponents barabar karo - done.*

QUESTION 103 | Difficult | AP from t4 and t9 (T9)

Q103. In an AP, the 4th term is 15 and the 9th term is 30. (i) Find a and d. (ii) Which term of the AP is 60?

SOLUTION — step by step

Step 1. (i) a + 3d = 15 and a + 8d = 30.

Step 2. Subtract: 5d = 15, d = 3; then a = 15 - 9 = 6.

Step 3. AP: 6, 9, 12, 15, ... with $t_n = 3n + 3$.

Step 4. (ii) 3n + 3 = 60 gives 3n = 57, n = 19.

Step 5. 60 is the 19th term.

» **Why this step?** The index gap (9 - 4 = 5) is exactly the coefficient of d after subtraction - 5d = 15.

Answer: (i) a = 6, d = 3 (ii) the 19th term.

Hinglish: *5d = 15 se d = 3, a = 6. 60 kaunsa term? 3n + 3 = 60, n = 19. Two-condition AP ka standard 4-step recipe - aur yahi 4 steps exam mein full marks dilate hain.*

QUESTION 104 | Difficult | Sierpinski area crossing 1/10 (T13)

Q104. Starting from area 1 at Stage 0, the black area of the Sierpiński triangle is $(3/4)^n$ at Stage n. Find the first stage at which the black area drops below 1/10 of the original.

SOLUTION — step by step

Step 1. Need $(3/4)^n < 0.1$.

Step 2. Compute powers: $(3/4)^7 = 2187/16384 \approx 0.1335$; $(3/4)^8 = 6561/65536 \approx 0.1001$ (just above 0.1!); $(3/4)^9 = 19683/262144 \approx 0.0751$.

Step 3. Stage 8 is a heartbreaker: 0.1001 is still (barely) above 1/10.

Step 4. The first stage strictly below 1/10 is Stage 9 (area ≈ 0.075).

» **Why this step?** 'Drops below' is strict - and the fraction 6561/65536 = 0.10011... shows Stage 8 misses by a whisker. Compute rather than eyeball.

Answer: Stage 9.

Hinglish: *Stage 8 pe area 0.1001 - bas 0.0001 se bach gaya! Stage 9 pe 0.075 - ab neeche. Fraction mein compare karo: 6561/65536 vs 1/10 = 6553.6/65536 - ab samajh aaya Stage 8 fail hua.*

Level 4 · Boss Zone: advanced + HOTS (Olympiad-lite):

QUESTION 105 | High | GP: sum 13/12, product -1 (T12)

Q105. The sum of the first three terms of a GP is 13/12 and their product is -1. Find the common ratio and the three terms.

SOLUTION — step by step

Step 1. Let the terms be a/r, a, ar. Product = (a/r) × a × (ar) = $a^3 = -1$, so a = -1.

Step 2. Sum: a(1/r + 1 + r) = 13/12, so -(1/r + 1 + r) = 13/12.

Step 3. Thus 1/r + r = -13/12 - 1 = -25/12.

Step 4. Multiply by 12r: $12 + 12r^2 = -25r$, i.e. $12r^2 + 25r + 12 = 0$.

Step 5. Split the middle term: $12r^2 + 16r + 9r + 12 = 4r(3r + 4) + 3(3r + 4) = (3r + 4)(4r + 3) = 0$.

Step 6. r = -4/3 or r = -3/4.

Step 7. For r = -3/4: terms 4/3, -1, 3/4 (sum 16/12 - 12/12 + 9/12 = 13/12 ✓, product -1 ✓).

Step 8. For r = -4/3: terms 3/4, -1, 4/3 - the same three numbers in reverse order.

» **Why this step?** Product of a/r × a × ar = a^3 - three symmetric terms reduce the product condition to a single cube, and the quadratic factorises by splitting.

Answer: r = -3/4 (or -4/3); terms are 4/3, -1, 3/4 in some order.

Hinglish: *Symmetric writing se product = a^3 = -1, matlab middle term -1. Quadratic $12r^2$ + 25r + 12 middle-term split se (3r + 4)(4r + 3). Do r aaye par dono same teen numbers dete hain, bas order ulta. Boss-level question - ye kar liya toh chapter ka raja tum ho.*

QUESTION 106 | High | Proof: t4, t10, t16 of a GP are in GP (T14)

Q106. If the 4th, 10th and 16th terms of a GP are x, y and z respectively, prove that x, y, z are in GP.

SOLUTION — step by step

Step 1. Let the GP have first term a and ratio r. Then $x = ar^3$, $y = ar^9$, $z = ar^{15}$.

Step 2. To show x, y, z are in GP it suffices to prove $y^2 = xz$.

Step 3. $y^2 = (ar^9)^2 = a^2r^{18}$.

Step 4. $xz = (ar^3)(ar^{15}) = a^2r^{18}$.

Step 5. $y^2 = xz$, hence x, y, z are in GP (with common ratio r^6).

» **Why this step?** Equally spaced terms of a GP (gap 6 here) always form a GP themselves - the middle term squared equals the product of the extremes.

Answer: Proved: $y^2 = a^2r^{18} = xz$, so x, y, z are in GP.

Hinglish: *Proof sirf $y^2 = xz$ dikhane ka hai. Exponents jodo: 3 + 15 = 18 = 9 + 9. Bas! GP ke equally spaced terms hamesha GP banate hain - middle wale ka square = dono bahar wale ka product.*

QUESTION 107 | High | GP: sum 26, sum of squares 364 (T12)

Q107. The sum of the first three terms of a GP is 26 and the sum of their squares is 364. Find the terms of the GP.

SOLUTION — step by step

Step 1. Write the terms as a/r, a, ar (three terms of a GP with middle a).

Step 2. Sum: $a(1/r + 1 + r) = 26$.

Step 3. Squares: $a^2(1/r^2 + 1 + r^2) = 364$.

Step 4. Hunt for a friendly GP by testing small integer GPs: 2, 6, 18 has sum 26 ✓ and squares 4 + 36 + 324 = 364 ✓.

Step 5. So a = 6, r = 3: terms 2, 6, 18. Reversing the order also works: 18, 6, 2 (r = 1/3).

Step 6. Verify no other simple GP fits: any other solution would need the same sum and square-sum, and 2, 6, 18 satisfies both exactly.

» **Why this step?** With a/r, a, ar the two conditions are two equations in a and r - but a quick scan for a 'sum 26' GP often cracks it faster than the algebra.

Answer: 2, 6, 18 (or 18, 6, 2).

Hinglish: *2 + 6 + 18 = 26 aur squares 4 + 36 + 324 = 364 - dono ek saath match! Isliye GP 2, 6, 18 (ya ulta 18, 6, 2). Aise double-condition GP questions mein pehle chhote integer GPs try karo - 2, 6, 18 jaisi list sabse pehle dimaag mein aani chahiye.*

QUESTION 108 | High | Self-summing recursion P-n (T5/T14)

Q108. Suppose $P_1 = 1$, $P_2 = 2$, and for $n > 2$, $P_n = P_1 + P_2 + \ldots + P_{n-1} + 1$. Find P_3 to P_8. Can you find a simpler recursive formula for P_n? And an explicit formula?

SOLUTION — step by step

Step 1. $P_3 = 1 + 2 + 1 = 4$.

Step 2. $P_4 = 1 + 2 + 4 + 1 = 8$; $P_5 = 1 + 2 + 4 + 8 + 1 = 16$.

Step 3. $P_6 = 31 + 1 = 32$; $P_7 = 63 + 1 = 64$; $P_8 = 127 + 1 = 128$.

Step 4. Values: 4, 8, 16, 32, 64, 128 - powers of 2 appearing!

Step 5. Simpler recursion: notice $P_n - 1$ = sum of all earlier terms, so P_{n+1} = (sum up to P_n) + P_n = (P_n - 1) + 1 + P_n = $2P_n$. So $P_n = 2P_{n-1}$.

Step 6. Explicit formula: $P_n = 2^{n-1}$ (check: $P_8 = 2^7 = 128$ ✓).

» **Why this step?** The 'sum of everything before me + 1' rule hides a doubling: the running total up to P_n is exactly P_n - 1, and adding P_n doubles it.

Answer: $P_3 \ldots P_8 = 4, 8, 16, 32, 64, 128$; simpler recursion $P_n = 2P_{n-1}$; explicit $P_n = 2^{n-1}$.

Hinglish: *4, 8, 16, 32, 64, 128 - GP mil gaya andar! Trick: P_n - 1 hi pichhla total hai, toh naya term = purana total + naya = double. Ye 'self-summing' recursion class ke best puzzles mein se hai.*

QUESTION 109 | High | Self-summing recursion W-n (T5/T6)

Q109. Suppose $W_1 = 1$, $W_2 = 2$, and for $n > 2$, $W_n = W_1 + W_2 + \ldots + W_{n-2} + 2$. Find W_1 through W_8. Do you recognise this sequence? Justify your recognition with a simpler recursion.

SOLUTION — step by step

Step 1. $W_3 = W_1 + 2 = 3$.

Step 2. $W_4 = W_1 + W_2 + 2 = 1 + 2 + 2 = 5$.

Step 3. $W_5 = 1 + 2 + 3 + 2 = 8$; $W_6 = 1 + 2 + 3 + 5 + 2 = 13$.

Step 4. $W_7 = 1 + 2 + 3 + 5 + 8 + 2 = 21$; $W_8 = 1 + 2 + 3 + 5 + 8 + 13 + 2 = 34$.

Step 5. Sequence: 1, 2, 3, 5, 8, 13, 21, 34 - the Virahānka-Fibonacci sequence!

Step 6. Why: the total up to W_{n-2} equals W_{n-1} - 2 (it is the full sum minus the last +2). So $W_n = (W_{n-1} - 2) + 2 = \ldots$ more precisely, compare W_n and W_{n-1}: $W_n - W_{n-1} = W_{n-2}$, giving $W_n = W_{n-1} + W_{n-2}$ for n at least 4 - exactly the Virahānka rule.

» **Why this step?** A 'sum up to n - 2, plus 2' rule secretly subtracts to a two-term recursion: consecutive W values differ by a W term.

Answer: 1, 2, 3, 5, 8, 13, 21, 34 - the Virahānka-Fibonacci sequence; indeed $W_n = W_{n-1} + W_{n-2}$.

Hinglish: *1, 2, 3, 5, 8, 13, 21, 34 - Virahānka-Fibonacci wapas! Dekho kaise: W_n minus W_{n-1} = W_{n-2} bachta hai. Poora sum likhne wala rule asal mein do-term recursion ka nakab hai.*

QUESTION 110 | High | Sum of first n odds = n squared (T10/T14)

Q110. Prove that $1 + 3 + 5 + ... + (2n - 1) = n^2$ in two different ways: (i) using the average × count idea, and (ii) using the natural-number sum formula.

SOLUTION — step by step

Step 1. (i) The n odd numbers are equally spaced from 1 to 2n - 1.

Step 2. Average = (1 + 2n - 1)/2 = n. Sum = average × count = n × n = n^2. ✓

Step 3. (ii) The odd numbers are the naturals up to 2n with the evens removed:

Step 4. Sum = $(1 + 2 + ... + 2n) - (2 + 4 + ... + 2n) = S_{2n} - 2S_n$.

Step 5. = $2n(2n + 1)/2 - 2 \times n(n + 1)/2 = n(2n + 1) - n(n + 1) = 2n^2 + n - n^2 - n = n^2$. ✓

» **Why this step?** Route (ii) shows the odd-sum identity is just $S_{2n} - 2S_n$: subtract the evens (each twice) from the naturals - pure formula algebra.

Answer: Both methods give n^2.

Hinglish: *Do proof, ek hi natija. Pehla: average n, count n, sum n^2. Doosra: 2n tak ke naturals minus do baar evens - algebra n^2 pe khatam. Dono yaad rakho, koi bhi exam maange ready ho.*

QUESTION 111 | High | Virahanka partial-sum identity (T6/T14)

Q111. For the Virahānka-Fibonacci sequence 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ... verify with n = 1, 2, 3, 4 that $V_1 + V_2 + ... + V_n = V_{n+2} - 2$, and explain why the identity keeps working for every n.

SOLUTION — step by step

Step 1. n = 1: sum = 1; $V_3 - 2 = 3 - 2 = 1$ ✓.

Step 2. n = 2: sum = 3; $V_4 - 2 = 5 - 2 = 3$ ✓.

Step 3. n = 3: sum = 6; $V_5 - 2 = 8 - 2 = 6$ ✓.

Step 4. n = 4: sum = 11; $V_6 - 2 = 13 - 2 = 11$ ✓.

Step 5. Why it works: the sum up to n equals $V_{n+2} - 2$ can be proved by telescoping - since $V_{k+2} = V_{k+1} + V_k$, each $V_k = V_{k+2} - V_{k+1}$; adding these for k = 1 to n makes the middle terms cancel, leaving $V_{n+2} - V_2 = V_{n+2} - 2$.

» **Why this step?** Rewriting $V_k = V_{k+2} - V_{k+1}$ turns every term into a difference of its neighbours - adding them up collapses everything except the two ends.

Answer: Verified for n = 1..4; the identity holds for all n by telescoping: the sum telescopes to $V_{n+2} - V_2 = V_{n+2} - 2$.

Hinglish: *1, 3, 6, 11 - sab $V_{n+2} - 2$ se match. Telescoping ka jaadu: har term ko do padosi terms ka difference likho, beech ka sab cancel. Ye identity MCQ aur one-liner proofs mein direct kaam aati hai.*

QUESTION 112 | High | AP transformations keep AP (T14)

Q112. $t_1, t_2, t_3, \ldots$ is an AP with first term a and common difference d. Show that each of these is also an AP, and find its first term and common difference: (i) $2t_1, 2t_2, 2t_3, \ldots$ (ii) $t_1 + 7, t_2 + 7, t_3 + 7, \ldots$ (iii) $3t_1 - 5, 3t_2 - 5, 3t_3 - 5, \ldots$

SOLUTION — step by step

Step 1. Recall $t_n = a + (n - 1)d$.

Step 2. (i) New term: $2t_n = 2a + (n - 1)(2d)$ - an AP with first term 2a and common difference 2d.

Step 3. (ii) New term: $t_n + 7 = (a + 7) + (n - 1)d$ - an AP with first term a + 7 and the SAME difference d.

Step 4. (iii) New term: $3t_n - 5 = (3a - 5) + (n - 1)(3d)$ - an AP with first term 3a - 5 and difference 3d.

Step 5. General law: multiplying every term by m multiplies d by m; adding a constant c shifts a but leaves d unchanged. Any combination $mt_n + c$ is again an AP.

» **Why this step?** The AP property lives entirely in the common difference - scaling scales it, shifting spares it. That is why linear edits of APs stay APs.

Answer: (i) AP, 2a and 2d (ii) AP, a + 7 and d (iii) AP, 3a - 5 and 3d.

Hinglish: *AP ka DNA uska d hai. 2 se multiply = d double. +7 = sirf shuruaat aage, d wahi. 3t - 5 = d triple, start shift. Formula mein n ka coefficient hi d hai - usse dekh ke turant likh do.*

QUESTION 113 | High | Triangles make squares (T14)

Q113. Let $T_n = n(n + 1)/2$ be the nth triangular number. Verify for n = 1, 2, 3, 5 that $8T_n + 1$ is a perfect square, then prove it algebraically and identify the square.

SOLUTION — step by step

Step 1. n = 1: $8(1) + 1 = 9 = 3^2$ ✓.

Step 2. n = 2: $8(3) + 1 = 25 = 5^2$ ✓.

Step 3. n = 3: $8(6) + 1 = 49 = 7^2$ ✓.

Step 4. n = 5: $8(15) + 1 = 121 = 11^2$ ✓.

Step 5. Proof: $8T_n + 1 = 8 \times n(n + 1)/2 + 1 = 4n^2 + 4n + 1 = (2n + 1)^2$ - always the square of the odd number 2n + 1.

» **Why this step?** $4n^2 + 4n + 1$ is a perfect-square expansion: $(2n + 1)^2 = 4n^2 + 4n + 1$. Eight triangles plus one centre dot arrange into a square of odd side.

Answer: Verified; algebraically $8T_n + 1 = (2n + 1)^2$.

Hinglish: *8 triangles + 1 dot = perfect square - geometrically bhi sach hai, algebraically bhi: $4n^2 + 4n + 1 = (2n + 1)^2$. Ye identity Pythagoras ke ancestor Baudhayana ke sulva-sutras mein bhi milti hai - triangular aur square numbers ka rishta.*

QUESTION 114 | High | Counting AP terms in a window (T8)

Q114. How many terms of the AP 9, 15, 21, 27, ... lie between 100 and 500?

SOLUTION — step by step

Step 1. $t_n = 9 + (n - 1) \times 6 = 6n + 3$.

Step 2. Need $100 < 6n + 3 < 500$.

Step 3. Lower: $6n > 97$, $n > 16.17$, so n at least 17 ($t_{17} = 105$ - the first term above 100).

Step 4. Upper: $6n < 497$, $n < 82.83$, so n at most 82 ($t_{82} = 495$ - the last term below 500).

Step 5. Count = 82 - 17 + 1 = 66 terms.

» **Why this step?** Inclusive counting of a whole-number range of n: (last n - first n) + 1. Compute the boundary terms to display them.

Answer: 66 terms.

Hinglish: *n ka window: 17 se 82. Count = 82 - 17 + 1 = 66. +1 bhoolna is question ki sabse common maut hai. Boundary terms (105 aur 495) likh do toh checker ko proof mil jaata hai.*

QUESTION 115 | High | Race between 2 to the n and 3n-1 (T14)

Q115. Two sequences run side by side: $t_n = 2^n$ and $s_n = 3n - 1$. List the first five terms of each. At which position does t_n first become strictly greater than s_n?

SOLUTION — step by step

Step 1. t: 2, 4, 8, 16, 32. s: 2, 5, 8, 11, 14.

Step 2. n = 1: 2 = 2 (tie). n = 2: 4 < 5 (s leads!).

Step 3. n = 3: 8 = 8 (tie again).

Step 4. n = 4: 16 > 11 - the power sequence overtakes for the first time.

Step 5. Answer: n = 4; and from here the gap only widens (32 vs 14, 64 vs 17, ...) because doubling beats adding 3 forever.

» **Why this step?** Exponential growth starts slow and even loses early laps, but once it passes any linear sequence it never looks back.

Answer: At n = 4 (16 > 11).

Hinglish: *Interesting race: 2 par tie, phir 5 ne 4 ko hara diya!, 8 par phir tie, aur 4th position pe 16 > 11 - ab GP hamesha ke liye aage. GP vs AP ki race ki yahi kahani hai - dheere shuru, phir bhagad billa.*

QUESTION 116 | High | Sum of first n even numbers (T10)

Q116. Prove that 2 + 4 + 6 + ... + 2n = n(n + 1). Verify it for n = 10 and connect the result to the odd-number sum.

SOLUTION — step by step

Step 1. Factor out 2: 2 + 4 + ... + 2n = 2(1 + 2 + ... + n) = 2 × n(n + 1)/2 = n(n + 1).

Step 2. Verify n = 10: 2 + 4 + ... + 20 = 2 × 55 = 110, and n(n + 1) = 10 × 11 = 110 ✓.

Step 3. Connection: sum of first n evens - sum of first n odds = n(n + 1) - n^2 = n. Makes sense: the k-th even exceeds the k-th odd by exactly 1, over n pairs that is n in total.

» **Why this step?** Evens are $2S_n$; odds are n^2; their difference n(n + 1) - n^2 = n counts the +1 gap in every pair.

Answer: Proved: the sum is n(n + 1); for n = 10 it equals 110.

Hinglish: *Even numbers = 2 × (1 se n tak). Seedha $2S_n$ = n(n + 1). Aur sundar connection: evens minus odds = n, kyunki har jodi mein sirf 1 ka farak hai.*

QUESTION 117 | HOTS | AP savings vs GP savings (T14)

Q117. Two friends save money for 12 weeks. Asha puts in ₹1 in week 1, ₹2 in week 2, ₹3 in week 3, and so on (increasing by ₹1 each week). Bhola puts in ₹1 in week 1, ₹2 in week 2, ₹4 in week 3, and so on (doubling each week). Who has saved more in total by the end of week 12, and by how much?

SOLUTION — step by step

Step 1. Asha's weekly deposits: 1, 2, 3, ..., 12 - total = 12 × 13/2 = ₹78.

Step 2. Bhola's weekly deposits: 1, 2, 4, 8, ..., 2^{11} - total = 1 + 2 + 4 + ... + 2048.

Step 3. Adding a power of 2 to the running total gives the NEXT power of 2 minus 1: 1 + 2 = 3, + 4 = 7, + 8 = 15, ... so the total is 2^{12} - 1 = 4096 - 1 = ₹4095.

Step 4. Bhola has saved 4095 - 78 = ₹4017 more.

Step 5. Lesson: an AP that adds ₹1 per week ends at 78; a GP that doubles ends past ₹4000 - multiplicative growth demolishes additive growth.

» **Why this step?** The doubling sum 1 + 2 + ... + 2^k = 2^{k+1} - 1 - each partial sum is one short of the next power of 2.

Answer: Bhola: ₹4095 vs Asha: ₹78 - Bhola leads by ₹4017.

Hinglish: *Asha total ₹78, Bhola ₹4095! Doubling ka sum hamesha agle power se ek kam hota hai - 4095 = 2^{12} - 1. GP ne AP ko 50 guna se zyada hara diya. Ye comparison exam ka favourite HOTS hai.*

QUESTION 118 | HOTS | Membership in 2 to the n plus 1 (T4/T14)

Q118. A sequence is given by $t_n = 2^n + 1$. (i) List the first five terms. (ii) Is 129 a term? (iii) Is 100 a term? Give full reasoning.

SOLUTION — step by step

Step 1. (i) $t_1 = 3$, $t_2 = 5$, $t_3 = 9$, $t_4 = 17$, $t_5 = 33$.

Step 2. (ii) Set $2^n + 1 = 129$, so $2^n = 128 = 2^7$, n = 7. Natural - yes, 129 is the 7th term.

Step 3. (iii) Set $2^n + 1 = 100$, so $2^n = 99$. But 99 is not a power of 2 (powers near it: 64 and 128), so no natural n exists.

Step 4. Hence 100 is NOT a term - even though 99 looks close to a power of 2, closeness is not membership.

» **Why this step?** Membership reduces to: is (k - 1) a power of 2? Exact powers only - 'almost' never counts.

Answer: (i) 3, 5, 9, 17, 33 (ii) yes - the 7th term (iii) no - 99 is not a power of 2.

Hinglish: *129 = 128 + 1 = 2^7 + 1 - 7th term. 100? Isse ek kam 99 - koi power of 2 nahi (64 ke baad seedha 128). GP-style membership mein exact chahiye, lagbhag nahi chalta.*

QUESTION 119 | HOTS | Frog crossing a pit (T13/T14)

Q119. A frog's first jump covers 1 m. Every later jump covers 3/4 of the distance of the jump before it. The frog must cross a pit 3.5 m wide (measured from the take-off edge; landing exactly at 3.5 m counts as crossing). After how many jumps has it covered at least 3.5 m in total?

SOLUTION — step by step

Step 1. Jump distances: 1, 3/4, 9/16, 27/64, ... - a GP with a = 1, r = 3/4.

Step 2. Running totals: after jump 1: 1. After jump 2: 1.75. Jump 3: + 0.5625 = 2.3125. Jump 4: + 0.421875 = 2.734375.

Step 3. Jump 5: + 0.31640625 = 3.05078125 (still short). Jump 6: + 0.2373046875 = 3.2880859375 (short).

Step 4. Jump 7: + 0.177978515625 = 3.466064453125 (short - just 3 cm to go!).

Step 5. Jump 8: + 0.13348388671875 = 3.59954833984375 > 3.5 - crossed on the 8th jump.

» **Why this step?** A shrinking GP's partial sums creep towards a limit (here 4 m) - so the frog eventually covers any distance below 4 m, but needs many tiny jumps near the end.

Answer: The frog crosses on the 8th jump (total 3.5995... m).

Hinglish: *Har jump 3/4 guna - totals 1, 1.75, 2.3125... 7th jump ke baad bhi 3.466 - sirf 3 cm reh gaye! 8th jump pe 3.599 - paar. Interesting: total kabhi 4 m se aage nahi jayega, limit hai. Decay GP ka sum limit tak ghisti chalti hai.*

QUESTION 120 | HOTS | Three-digit numbers with digits in AP (T14)

Q120. How many three-digit numbers between 100 and 200 have their digits in arithmetic progression (reading left to right)? List them all.

SOLUTION — step by step

Step 1. Numbers look like 1, 1 + d, 1 + 2d where d is the common difference of the digits (d an integer).

Step 2. Each digit must be between 0 and 9: 1 + 2d at least 0 gives d at least -0.5, so d at least 0 (integer); 1 + 2d at most 9 gives d at most 4.

Step 3. So d = 0, 1, 2, 3, 4.

Step 4. d = 0: 111. d = 1: 123. d = 2: 135. d = 3: 147. d = 4: 159.

Step 5. There are exactly 5 such numbers: 111, 123, 135, 147, 159. (Descending digit patterns like 210 start with 2, outside 100-200.)

» Why this step? Fix the first digit, then the digit-AP condition forces the shape 1, 1 + d, 1 + 2d - the bounds 0 to 9 clamp d to exactly five values.

Answer: Exactly 5: 111, 123, 135, 147, 159.

Hinglish: *Digits AP mein: 1, 1+d, 1+2d. 0-9 ki boundary ne d ko 0 se 4 tak simit kar diya - 5 numbers: 111, 123, 135, 147, 159. Aur sundar fact: aise har 3-digit number ka digit-sum 3 ka multiple hota hai (middle digit × 3) - isliye ye 3 se divisible hote hain!*

8.5 Rapid MCQs: Choose the Correct Option (20)

Rapid-fire round: 4 options, ek hi correct. Pehle khud choose karo, phir colored Answer strip se check karo. Har galat option ka "Why wrong" bhi diya hai - taki exam me wahi trap tumhe na phasaye.

MCQ 1 | Normal | Identify the AP

1. Which of the following is an arithmetic progression?

A. 1, 2, 4, 8, 16

B. 2, 5, 8, 11, 14

C. 1, 4, 9, 16, 25

D. 1, 1/2, 1/4, 1/8

Correct Answer: (B)

2, 5, 8, 11, 14 has a constant difference of 3 between consecutive terms - the definition of an AP. The others are a GP (doubling), the squares (differences 3, 5, 7 grow) and a halving GP.

Why (A) is wrong: That doubles each time - a GP with r = 2.

Why (C) is wrong: Differences are 3, 5, 7 - growing, not constant.

Why (D) is wrong: Each term halves - a GP with r = 1/2.

Hinglish: *AP ka test = constant ADD. Option B mein har baar +3. Baaki teeno multiplication wale ya growing-gap wale - GP ya neither.*

MCQ 2 | Normal | Common difference

2. The common difference of the AP 5, 9, 13, 17, ... is:

A. 4

B. 5

C. 9

D. -4

Correct Answer: (A)

$d = t_2 - t_1 = 9 - 5 = 4$ (and 13 - 9 = 4 confirms it).

Why (B) is wrong: 5 is the first term a, not the difference.

Why (C) is wrong: 9 is the second term, not a difference.

Why (D) is wrong: The AP increases, so d must be positive.

Hinglish: *Do consecutive terms ka farak hi d hai: 9 - 5 = 4. Ek baar doosre pair se bhi check karo - 13 - 9 = 4 ✓.*

MCQ 3 | Normal | Tenth term of an AP

3. The 10th term of the AP 7, 11, 15, 19, ... is:

A. 43

B. 47

C. 39

D. 51

Correct Answer: (A)

$a = 7$, $d = 4$. $t_{10} = 7 + 9 \times 4 = 7 + 36 = 43$.

Why (B) is wrong: Used 10 steps instead of 9: 7 + 40 = 47.

Why (C) is wrong: Forgot one step: 7 + 8 × 4 = 39.

Why (D) is wrong: Overshot the arithmetic: wrong step count.

Hinglish: *7 + 9 steps × 4 = 43. Yaad rakho (n - 1) multiply hota hai, n nahi - yahi galti option C jaise traps banti hai.*

MCQ 4 | Normal | Which term is 109

4. Which term of the AP 4, 9, 14, 19, ... is 109?

A. 21st

B. 22nd

C. 20th

D. 23rd

Correct Answer: (B)

a = 4, d = 5. Solve 4 + (n - 1) × 5 = 109: (n - 1) × 5 = 105, n - 1 = 21, n = 22.

Why (A) is wrong: 21 is the number of steps, not the position.

Why (C) is wrong: Miscounted the division: 105/5 = 21.

Why (D) is wrong: Added two extra somewhere - check (n - 1) × 5 = 105.

Hinglish: *(109 - 4)/5 = 21 steps, toh 22nd term. Steps + 1 = position - off-by-one ka sawal hai.*

MCQ 5 | Normal | Sum of first 15 naturals

5. The sum 1 + 2 + 3 + ... + 15 equals:

A. 120

B. 105

C. 240

D. 135

Correct Answer: (A)

$S_{15} = 15 \times 16/2 = 15 \times 8 = 120$.

Why (B) is wrong: That is S_{14} - one term short.

Why (C) is wrong: Forgot to halve: 15 × 16 = 240.

Why (D) is wrong: Wrong pairing - recompute 15 × 8.

Hinglish: *n(n+1)/2 = 15 × 16/2. 16 ka aadha 8, 15 × 8 = 120. Ratta nahi - half karo pehle, multiply easy ho jaata hai.*

MCQ 6 | Normal | Triangular number

6. The 8th triangular number is:

A. 28

B. 36

C. 45

D. 24

Correct Answer: (B)

$T_8 = 8 \times 9/2 = 36$ (also $1 + 2 + ... + 8$).

Why (A) is wrong: That is the 7th triangular number.

Why (C) is wrong: That is the 9th.

Why (D) is wrong: Used 8×6 - wrong formula.

Hinglish: *Triangular = S_n. $8 \times 9/2 = 36$. List yaad ho toh 1, 3, 6, 10, 15, 21, 28, 36 - aathwaan 36.*

MCQ 7 | Normal | Common ratio with sign

7. The common ratio of the GP 2, -6, 18, -54, ... is:

A. 3

B. -3

C. -2

D. 1/3

Correct Answer: (B)

$r = t_2/t_1 = -6/2 = -3$. Check: $18/(-6) = -3$ ✓.

Why (A) is wrong: Ignored the minus sign of the second term.

Why (C) is wrong: Divided the wrong pair.

Why (D) is wrong: Inverted the division - that is 1/r territory.

Hinglish: *Signs baari-baari palat rahe hain - matlab r negative. -6/2 = -3. Negative ratio = alternating GP.*

MCQ 8 | Normal | Fifth term of a GP

8. The 5th term of the GP 3, 12, 48, ... is:

A. 192

B. 768

C. 144

D. 576

Correct Answer: (B)

$a = 3$, $r = 4$. The 5th term is $ar^4 = 3 \times 4^4 = 3 \times 256 = 768$. (The exponent is 4, not 5 - the first term uses r^0.)

Why (A) is wrong: That is 3×4^3 - the 4th term, one step short.

Why (C) is wrong: Wrong ratio used (3 instead of 4).

Why (D) is wrong: Miscounted the exponent chain.

Hinglish: *$t_5 = ar^4$ - pehla term r^0 leta hai. $4^4 = 256$, $\times 3 = 768$. Exponent = position - 1.*

MCQ 9 | Normal | Virahanka next term

9. The next term of the sequence 1, 2, 3, 5, 8, 13, ... is:

A. 18

B. 21

C. 26

D. 16

Correct Answer: (B)

Each term is the sum of the previous two (Virahānka-Fibonacci): 8 + 13 = 21.

Why (A) is wrong: Added 13 + 5 - skipped a term.

Why (C) is wrong: Doubled 13 - that is a GP rule, not this one.

Why (D) is wrong: Added 13 + 3 - wrong window.

Hinglish: *Do pichhle ka sum: 8 + 13 = 21. Virahānka sequence - 1, 2, 3, 5, 8, 13, 21, 34, 55, 89 ratt lo.*

MCQ 10 | Normal | Recursive rule to terms

10. A sequence has $t_1 = 2$ and $t_n = t_{n-1} + 5$. Its 6th term is:

A. 27

B. 32

C. 30

D. 22

Correct Answer: (A)

This is an AP with a = 2, d = 5: $t_6 = 2 + 5 \times 5 = 27$.

Why (B) is wrong: Used 6 steps: 2 + 30.

Why (C) is wrong: Started the count at 0 and slipped.

Why (D) is wrong: Subtracted instead of added.

Hinglish: *+5 ka AP. 6th term = 2 + (6-1) × 5 = 27. Five steps from the start - not six.*

MCQ 11 | Normal | Halving GP next term

11. The sequence 1, 1/2, 1/4, 1/8, ... continues with next term:

A. 1/16

B. 1/10

C. 1/12

D. 1/6

Correct Answer: (A)

Common ratio r = 1/2, so the next term is 1/8 × 1/2 = 1/16.

Why (B) is wrong: Subtracted a constant - the pattern multiplies.

Why (C) is wrong: Wrong halving anchor.

Why (D) is wrong: Mixed the addition and division patterns.

Hinglish: *Har term aadhi - r = 1/2. 1/8 ka aadha 1/16. Halving GP kabhi zero par nahi rukti, bas girti jaati hai.*

MCQ 12 | Normal | Sierpinski stage 4 count

12. In the Sierpinski triangle (Stage 0 = 1 black triangle), the number of black triangles at Stage 4 is:

A. 27

B. 81

C. 243

D. 12

Correct Answer: (B)

The count is 3^n at Stage n: Stage 4 gives $3^4 = 81$.

Why (A) is wrong: That is Stage 3 - the classic off-by-one.

Why (C) is wrong: That is Stage 5.

Why (D) is wrong: Added 3 each stage instead of multiplying.

Hinglish: *Stage 0 se shuru - count 3^n. Stage 4 = 3^4 = 81. Off-by-one ka trap: 27 Stage 3 hai.*

MCQ 13 | Normal | Graph of an AP

13. When the pairs (n, t_n) of an arithmetic progression are plotted, the points:

A. lie on a straight line

B. lie on a curve rising ever more steeply

C. form a zigzag with no pattern

D. lie on a circle

Correct Answer: (A)

$t_n = a + (n - 1)d$ is linear in n with slope d - equal steps in n give equal steps in value, hence a straight line.

Why (B) is wrong: That is the GP graph - equal ratios bend the curve.

Why (C) is wrong: There is a perfect pattern - linear.

Why (D) is wrong: Nothing circular here.

Hinglish: *Constant difference = constant slope = seedhi line. Curve wali baat GP ke liye hoti hai.*

MCQ 14 | Normal | Graph of a GP

14. Plotting the GP 3, 6, 12, 24, 48 as pairs (1, 3), (2, 6), (3, 12), (4, 24), (5, 48) gives points that:

A. lie on one straight line

B. do not lie on a line - the rise steepens

C. alternate above and below the x-axis

D. all have the same y-value

Correct Answer: (B)

Slopes between consecutive points are 3, 6, 12, 24 - doubling, so the points bend upward on an exponential-style curve.

Why (A) is wrong: Equal slopes would be needed - they double instead.

Why (C) is wrong: All values are positive - no alternation.

Why (D) is wrong: The values double - clearly not constant.

Hinglish: *Slopes 3, 6, 12, 24 - badalte ja rahe hain, toh line nahi. GP ka graph tez-hoti curve hai.*

MCQ 15 | Normal | Zero as a term

15. Is 0 a term of the AP 5, 11, 17, 23, ...?

A. No - solving gives a fractional position

B. Yes - the 0th term

C. Yes - the 1st term

D. Cannot be decided

Correct Answer: (A)

$5 + 6(n - 1) = 0$ gives $n - 1 = -5/6$, so $n = 1/6$ - not a natural number. Hence 0 is not a term (the AP is positive and climbing away from 0).

Why (B) is wrong: Positions start at 1, not 0.

Why (C) is wrong: The first term is 5.

Why (D) is wrong: The equation decides it completely.

Hinglish: *$5 + 6(n-1) = 0$ se n aadha-aadha nikla - position exist nahi karti. Ye AP 5 se upar jaati hai, zero toh peechhe chhoot gaya.*

MCQ 16 | Normal | Meaning of negative d

16. If the common difference of an AP is negative, the AP:

A. decreases

B. increases

C. stays constant

D. alternates in sign

Correct Answer: (A)

Each term is d less than the previous one - a negative d means every step goes down (e.g. 11, 7, 3, -1 with d = -4).

Why (B) is wrong: Positive d does that.

Why (C) is wrong: Only d = 0 keeps terms constant.

Why (D) is wrong: Sign alternation needs a negative RATIO, not difference.

Hinglish: *d = step size with sign. Negative d = neeche utarna. Alternation GP ka (r negative) kaam hai, AP ka nahi.*

MCQ 17 | Normal | Taxi fare model

17. A taxi charges a fixed booking fee of ₹200 plus ₹40 per km. The total fare after n km is modelled by the sequence of fares for 1 km, 2 km, ... The nth fare equals:

A. 200 + 40n

B. 200 + 40(n - 1)

C. 40 + 200n

D. 200 × 40n

Correct Answer: (A)

Fares: 240, 280, 320, ... - an AP with a = 240, d = 40. Fare after n km = 240 + (n - 1) × 40 = 200 + 40n (equivalently: fee 200 plus 40 per km for n km).

Why (B) is wrong: That undercounts the first km's charge (gives 200 at n = 0).

Why (C) is wrong: Swapped the fixed fee and the rate.

Why (D) is wrong: Multiplying fee and rate is meaningless here.

Hinglish: *Fixed 200 + per-km 40 × n. Pehle km ke baad fare 240 hai - formula 200 + 40n dono taraf se match karta hai.*

MCQ 18 | Normal | Whose nth term is n(n+1)/2

18. The sequence whose nth term is n(n + 1)/2 is the sequence of:

A. triangular numbers

B. square numbers

C. odd numbers

D. powers of 2

Correct Answer: (A)

n(n + 1)/2 = 1 + 2 + ... + n - the triangular numbers 1, 3, 6, 10, 15, ...

Why (B) is wrong: Squares are n^2.

Why (C) is wrong: Odds are 2n - 1.

Why (D) is wrong: Powers of 2 are 2^n (or 2^{n-1}).

Hinglish: *n(n+1)/2 = 1 se n tak ka sum = triangular. Squares ka rule n^2 hai, odds ka 2n - 1, powers ka 2^n.*

MCQ 19 | Normal | Gauss classic

19. The value of 1 + 2 + 3 + ... + 100 is:

A. 5050

B. 5000

C. 10100

D. 4950

Correct Answer: (A)

$S_{100} = 100 \times 101/2 = 50 \times 101 = 5050$ - Gauss's famous schoolboy answer.

Why (B) is wrong: Rounded the pairing carelessly.

Why (C) is wrong: Forgot to halve 100 × 101.

Why (D) is wrong: That is S_{99}.

Hinglish: *100 × 101/2: aadha karo (50), 50 × 101 = 5050. Ye number toh ratta banta hai.*

MCQ 20 | Normal | Sierpinski area fate

20. The black area of the Sierpinski triangle at Stage n is $(3/4)^n$. As the stage number n grows larger and larger, this area:

A. approaches 0

B. approaches 1

C. stays equal to 1

D. grows without bound

Correct Answer: (A)

Multiplying by 3/4 again and again keeps shrinking the value: 1, 0.75, 0.5625, 0.42, ... - it tends to 0 even though it never exactly reaches it.

Why (B) is wrong: 1 is the Stage-0 value, not the destination.

Why (C) is wrong: Only Stage 0 has area 1.

Why (D) is wrong: Growth belongs to the triangle COUNT (3^n), not the area.

Hinglish: *Har baar 3/4 - value zero ki taraf ghisti jaati hai, par kabhi exactly zero nahi hoti. Count badhta hai (3^n), area ghatta hai - fractal ka paradox.*

8.6 Choose the Correct Round: Theory & Speed (20)

Doosra rapid round - har MCQ ka style: statement padho, correct option turant chuno. Ye round theory, definitions aur quick solving ko test karta hai. Answer strip ke neeche har galat option ka "Why wrong" bhi hai - taki exam ka trap tumhe na phasaye.

CTC 1 | Normal | AP definition

1. A list of numbers in which each term after the first is obtained by adding a fixed number to the previous term is called:

A. an arithmetic progression

B. a geometric progression

C. a recursive sequence

D. a finite series

Correct Answer: (A)

Constant ADDITION of a fixed number d defines the arithmetic progression.

Why (B) is wrong: Multiplication defines a GP.

Why (C) is wrong: 'Recursive' describes the rule style, not this defining property.

Why (D) is wrong: A series is a SUM, not a list.

Hinglish: *Add karte raho fixed number - AP. Multiply karte raho - GP. Naam hi kaafi hai is definition mein.*

CTC 2 | Normal | General form of an AP

2. The general form of an AP with first term a and common difference d is:

A. a, a + d, a + 2d, a + 3d, ...

B. a, ad, ad^2, ad^3, ...

C. a, a + d, a + 3d, a + 7d, ...

D. a, 2a, 3a, 4a, ...

Correct Answer: (A)

Term k (position k) equals a + (k - 1)d - so the list runs a, a + d, a + 2d, a + 3d, ...

Why (B) is wrong: That is the general form of a GP.

Why (C) is wrong: Gaps 1, 2, 4 grow - not constant.

Why (D) is wrong: That is a special list with ratio in addition form, not the general AP.

Hinglish: *Har term mein ek aur d judta hai - option A. Option B toh GP hai.*

CTC 3 | Normal | nth term formula of AP

3. The nth term of an AP is:

A. a + (n - 1)d

B. a + nd

C. a - (n - 1)d

D. n(a + d)

Correct Answer: (A)

From a to the nth term you take (n - 1) steps of size d: $t_n = a + (n - 1)d$.

Why (B) is wrong: Gives a + d at n = 1 - wrong first term.

Why (C) is wrong: Only for decreasing APs with |d| written positive.

Why (D) is wrong: No such formula exists.

Hinglish: *Ek kam steps - (n-1)d. n = 1 par formula a hi de - option A hi aisa karta hai.*

CTC 4 | Normal | Fractional ratio reading

4. The common ratio of the GP 5, 15/4, 45/16, ... is:

A. 3/4

B. 4/3

C. 3/5

D. 1/4

Correct Answer: (A)

r = (15/4)/5 = 15/20 = 3/4 (dividing by 5 multiplies by 1/5).

Why (B) is wrong: Inverted the fraction - that would make the list grow.

Why (C) is wrong: Divided 15 by 5 forgetting the /4.

Why (D) is wrong: Read the second term as 15, not 15/4.

Hinglish: *(15/4) ÷ 5 = 15/4 × 1/5 = 3/4. Fraction divide = ulta multiply.*

CTC 5 | Normal | Meaning of recursive rule

5. A recursive rule for a sequence is one that:

A. gives each term using previous terms

B. gives each term directly from its position n

C. uses only the first term forever

D. requires knowing all terms at once

Correct Answer: (A)

Recursive = build each new term from earlier term(s) plus a starting value - e.g. $t_n = t_{n-1} + 3$.

Why (B) is wrong: That is an explicit rule.

Why (C) is wrong: You need the recipe too, not just t_1.

Why (D) is wrong: You only need the previous terms, never the whole list.

Hinglish: *Recursive = pichhle terms se naya term banao. Direct-from-n wala explicit hota hai.*

CTC 6 | Normal | Meaning of explicit rule

6. The main advantage of an explicit rule like $u_n = 2n - 1$ is that:

A. any term can be found directly without earlier terms

B. it lists every term automatically

C. it works only for increasing sequences

D. it avoids all arithmetic

Correct Answer: (A)

Substitute the position n and the term appears - even u_{1170} needs one substitution, no earlier terms.

Why (B) is wrong: It generates any single term, not an automatic list.

Why (C) is wrong: $u_n = 2 - 5n$ is explicit and decreasing.

Why (D) is wrong: Arithmetic is still needed - just less of it.

Hinglish: *Explicit rule ka fayda: 1170th term bhi ek substitution mein. Listing ki zaroorat hi nahi.*

CTC 7 | Normal | Virahanka recursion

7. The Virahānka-Fibonacci sequence is defined by the rule:

A. $V_n = V_{n-1} + V_{n-2}$

B. $V_n = 2V_{n-1}$

C. $V_n = V_{n-1} + 1$

D. $V_n = n(n + 1)/2$

Correct Answer: (A)

Each term is the sum of the previous TWO terms, starting 1, 2 - giving 1, 2, 3, 5, 8, 13, ...

Why (B) is wrong: Doubling gives 1, 2, 4, 8 - a GP.

Why (C) is wrong: That gives naturals.

Why (D) is wrong: That is the triangular-number formula.

Hinglish: *Do pichhle ka jod - yahi Virahānka rule. Start 1, 2 (Fibonacci ke apne version mein 1, 1 hota hai).*

CTC 8 | Normal | Triangular nth term

8. The nth triangular number equals:

A. $n(n + 1)/2$

B. $n(n - 1)/2$

C. n^2

D. $2n - 1$

Correct Answer: (A)

$T_n = 1 + 2 + ... + n = n(n + 1)/2$.

Why (B) is wrong: Gives 10 at n = 5 - that is S_{n-1}.

Why (C) is wrong: That is squares.

Why (D) is wrong: That is odd numbers.

Hinglish: *Triangular = $S_n = n(n+1)/2$. n=5 pe 15 deta hai - check kar lo.*

CTC 9 | Normal | nth term of GP

9. The nth term of a GP with first term a and ratio r is:

A. ar^{n-1}

B. ar^n

C. $a + (n - 1)r$

D. arn

Correct Answer: (A)

First term = ar^0, second = ar^1, ..., nth = ar^{n-1} - the exponent is one less than the position.

Why (B) is wrong: Gives ar at n = 1 - skips the plain a.

Why (C) is wrong: That is an AP-style formula.

Why (D) is wrong: Not a formula - just letters.

Hinglish: *Exponent = n - 1. Pehla term r^0 wala - isliye option A.*

CTC 10 | Normal | Finite GP spotting

10. The sequence 6, 12, 24, 48, 96 is:

A. a finite GP with r = 2

B. an infinite GP

C. an AP with d = 6

D. an AP with d = 2

Correct Answer: (A)

Each term doubles (r = 2) and the list stops after 5 terms - a finite geometric progression.

Why (B) is wrong: The list stops - it is finite.

Why (C) is wrong: Differences 6, 12, 24 grow - not an AP.

Why (D) is wrong: Same reason - differences are not constant.

Hinglish: *Ratio constant 2, aur list ruk gayi - finite GP. Ye PDF ka hi favourite example hai.*

CTC 11 | Normal | Sierpinski carpet count

11. At Stage n of the Sierpiński square carpet (Stage 0 = full square), the number of small red squares is:

A. 8^n

B. 9^n

C. 3^n

D. n^8

Correct Answer: (A)

Each red square spawns 8 children (the centre ninth is removed): 1, 8, 64, 512, ... = 8^n at Stage n.

Why (B) is wrong: 9 pieces exist, but the centre one is removed.

Why (C) is wrong: 3^n belongs to the Sierpiński TRIANGLE.

Why (D) is wrong: Exponent and base are swapped.

Hinglish: *9 mein se 1 kata - 8 bachche. Har stage ×8: 8^n. 9^n hota agar kuch na kata hota.*

CTC 12 | Normal | Bouncing ball model

12. A ball bounces to 3/4 of its previous height each time. The bounce heights form:

A. a GP with r = 3/4 (shrinking)

B. an AP with d = -3/4

C. a GP with r = 4/3 (growing)

D. an AP with d = 3/4

Correct Answer: (A)

Each height MULTIPLIES the previous by 3/4: 18, 13.5, 10.125, ... - a geometric progression with ratio less than 1 (decay).

Why (B) is wrong: The change is proportional, not a fixed subtraction.

Why (C) is wrong: The ball never bounces HIGHER than before.

Why (D) is wrong: Same mix-up - it is multiplication, not addition.

Hinglish: *Multiply by 3/4 = GP, aur r < 1 toh shrinking. d wali baat tab hoti jab '18 cm kam' hota har baar.*

CTC 13 | Normal | Sum of first n odds

13. The sum of the first n odd numbers 1 + 3 + 5 + ... + (2n - 1) equals:

A. n^2

B. 2n - 1

C. n(n + 1)/2

D. n(n + 1)

Correct Answer: (A)

The average of the n odds is n and there are n of them: sum = n × n = n^2 (1 + 3 + 5 + 7 = 16 = 4^2).

Why (B) is wrong: That is the LAST odd number, not the sum.

Why (C) is wrong: That is the triangular sum.

Why (D) is wrong: That is the even-number sum.

Hinglish: *First n odds ka sum = n^2. Isi wajah se squares ke differences odd numbers hote hain - ulta connection.*

CTC 14 | Normal | Position of 137

14. In the sequence of odd numbers 1, 3, 5, 7, ..., the number 137 occupies position:

A. 69

B. 68

C. 137

D. 70

Correct Answer: (A)

Solve 2n - 1 = 137: n = 138/2 = 69.

Why (B) is wrong: Forgot the +1 before halving.

Why (C) is wrong: That is the value, not the position.

Why (D) is wrong: Off-by-one in the division.

Hinglish: *(137 + 1)/2 = 69 - odd number ka position = (value + 1)/2.*

CTC 15 | Normal | Hundredth term quick fire

15. If $t_n = 5n - 2$, then t_{100} equals:

A. 498

B. 500

C. 502

D. 480

Correct Answer: (A)

$5 \times 100 - 2 = 498$.

Why (B) is wrong: Forgot the -2.

Why (C) is wrong: Added 2 instead.

Why (D) is wrong: Arithmetic slip.

Hinglish: *Seedha substitute: 500 - 2 = 498. Ek second ka question hai ye.*

CTC 16 | Normal | d of a decreasing AP

16. The common difference of the AP 11, 7, 3, -1, -5, ... is:

A. -4

B. 4

C. -11

D. 0

Correct Answer: (A)

d = 7 - 11 = -4 (negative, because the AP falls).

Why (B) is wrong: That is the SIZE of the step, missing the sign.

Why (C) is wrong: That is the first term.

Why (D) is wrong: The terms clearly change.

Hinglish: *Gir rahi hai toh d negative: 7 - 11 = -4. Sign dikhana zaroori hai.*

CTC 17 | Normal | Membership test protocol

17. To test whether a number k is a term of the sequence $t_n = 3n + 4$, the correct procedure is:

A. solve $3n + 4 = k$ and accept only if n is a natural number

B. accept if k is odd

C. accept if k is positive

D. list terms until k appears or give up

Correct Answer: (A)

Solve for n: if n comes out a natural number, k is that term; a fractional n means k is not a term.

Why (B) is wrong: Parity decides nothing here (e.g. 9 is a term, 11 is not - both odd).

Why (C) is wrong: Positivity is far too weak.

Why (D) is wrong: Listing works but never proves absence - the equation does.

Hinglish: *Equation solve karo, n whole number? Haan toh member. Fraction? Reject. Odd/even waale shortcuts fail ho jaate hain.*

CTC 18 | Normal | Sierpinski triangle count

18. The number of black triangles at Stage n of the Sierpiński triangle (Stage 0 = 1 triangle) is:

A. 3^n

B. 3^{n-1}

C. n^3

D. 3n

Correct Answer: (A)

Stage 0 has $3^0 = 1$, Stage 1 has 3, Stage 2 has 9 - the exponent equals the stage number.

Why (B) is wrong: That would give 1 at Stage 1 - wrong.

Why (C) is wrong: Swapped base and exponent.

Why (D) is wrong: That is linear growth; fractals multiply.

Hinglish: *Stage 0 se shuru - isliye exponent n hai, n-1 nahi. Stage 3 pe 27 = 3^3.*

CTC 19 | Normal | Gauss pairing idea

19. The pairing proof of $S_n = 1 + 2 + ... + n$ works by:

A. writing the sum forwards and backwards and adding - each column sums to n + 1

B. multiplying all terms together

C. subtracting 1 from each term

D. counting only the even numbers

Correct Answer: (A)

S plus reversed-S gives 2S = n(n + 1) - every pair (k, n + 1 - k) adds to n + 1, and there are n columns.

Why (B) is wrong: Multiplication has no role in this proof.

Why (C) is wrong: Shifting terms does not pair anything.

Why (D) is wrong: Evens alone give a different sum.

Hinglish: *Aage-peechhe likho, jodo - har column n + 1. 2S = n(n + 1), isliye S = n(n+1)/2. Gauss aur Āryabhaṭa dono ka favourite.*

CTC 20 | Normal | Differences of squares

20. The differences between consecutive square numbers 1, 4, 9, 16, 25, ... are:

A. the consecutive odd numbers 3, 5, 7, 9, ...

B. the consecutive even numbers 2, 4, 6, ...

C. always 2

D. the prime numbers

Correct Answer: (A)

4 - 1 = 3, 9 - 4 = 5, 16 - 9 = 7, 25 - 16 = 9 - consecutive odds, because $(n + 1)^2 - n^2 = 2n + 1$.

Why (B) is wrong: Evens belong to the even-sum identity.

Why (C) is wrong: The differences grow.

Why (D) is wrong: Primes have no such role here.

Hinglish: *$(n+1)^2 - n^2 = 2n + 1$ - odd numbers nikalte hain. Isi se squares = odds ka running total.*